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Convergent information flows explain recurring firing patterns in cerebral cortex

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Cortical population events, short-lived patterns of neuronal activity that recur with consistency, are central to sensorimotor coordination. These reproducible firing patterns are often attributed to attractor dynamics, supported by strong mutual connectivity. However, by using multimodal datasets—including two-photon imaging, electrophysiology and electron microscopy—we show that these reproducible patterns do not involve strongly interconnected neurons. Instead, we show that cortical networks exhibit hierarchical modularity, with core neurons serving as high-information-flow nodes at module interfaces. These cores funnel activity but lack the structural signatures of pattern-completion units that are typically found in attractor networks. Using computational models, we find that distance-dependent connectivity is necessary and sufficient to produce the modularity and transient reproducible events observed in cortex. Our findings suggest that cortical networks are preconfigured to support sensorimotor coordination. This work redefines the structural and dynamical basis of cortical activity, with a focus on the relationship between modular structure and function.

All animals produce sensorimotor coordination to survive and thrive. At the top of the nervous system hierarchy, cortical population events of temporally correlated neuronal firing[1] orchestrate this coordination. Cortical events occur already before birth[2], promote specific behaviors[3,4], recur under similar stimulus conditions[5,6], retain their neuronal composition when reoccurring spontaneously[7,8] and can also be induced by repetitive coactivation[3,8]. A defining characteristic of cortical events is that they exhibit reproducible firing patterns in the participating cells[7,8]. This reproducibility is often viewed as evidence that these firing patterns are the dynamical signature of fixed point[9,10], limit cycle[11–13] or continuous[14–18] attractor networks. Attractor networks are powerful for understanding how neural activity can stabilize into specific, reproducible patterns. Neurons within an attractor network tend to excite each other with strong recursive mutual connections, creating a positive feedback that stabilizes the network's activity pattern[18–20]. In limit-cycle attractor models, strong mutual connections ensure periodic behavior by supporting both excitatory and inhibitory feedback, which drives the cycle forward and regulates its periodicity[21,22]. In continuous attractors, strong mutual connections maintain stability across a continuum of states, resisting perturbations and supporting stable representations of continuous variables like spatial position or orientation[15,17,23]. In the present work, we analyzed several datasets to assess whether reproducible cortical population events bear the structural and dynamical hallmarks of attractor networks, or whether an alternative mechanism better accounts for their emergence.

Results

Cortical population events, also known as ensemble, neural assembly and cell assembly events, are transiently occurring spiking patterns of groups of neurons[7,24,25]. The study of these events involves primarily the following two in vivo techniques: two-photon calcium imaging and electrophysiology[26]. Two-photon imaging enables simultaneous, high spatial-resolution recording of neural activity. Electrophysiological recordings provide high-temporal-resolution data on neuronal firing. To study cortical population events, we integrated multiple publicly available datasets—two-photon datasets from the MICrONS project[27–29] (mouse primary visual cortex, VISp) and equivalent experiments from the Allen Brain Observatory[30], the Goard lab[31] (mouse retrosplenial

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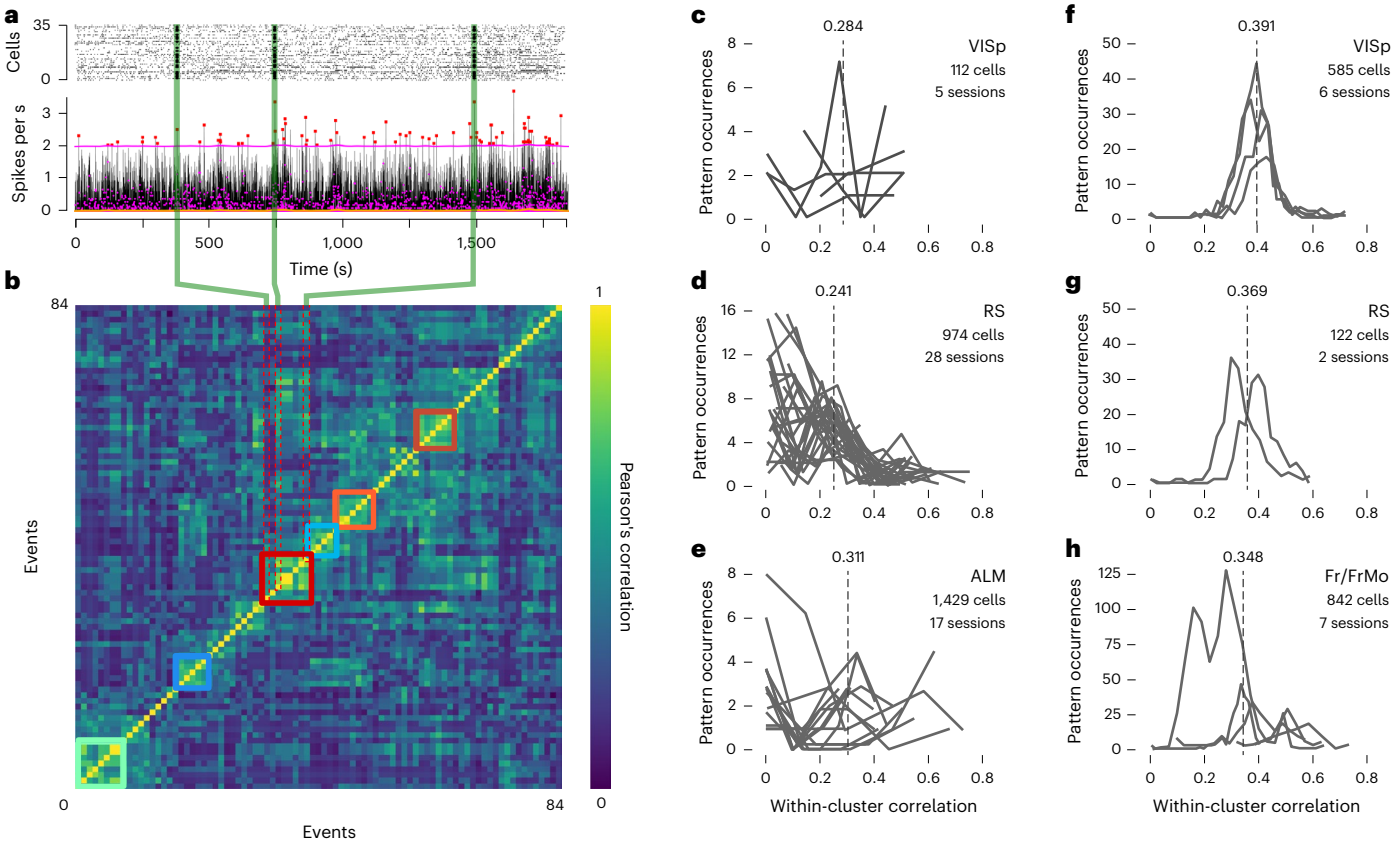

Fig. 1 | Reproducible activity patterns occur throughout cortex. a, Top, example calcium spikes raster plot from one MICrONS dataset (five available in total). High spike frequency indicates synchronous population events (for example, green shading). Bottom, population events were quantified as time intervals where the summed firing rate (black) exceeded a surrogate-based threshold (magenta) based on the baseline (orange). Among all the subthreshold minima (magenta points), those before and after each peak (red squares) are the beginning and end of each event. Each event was assigned a population binary vector (top, participating cells as black dots). **b**, Population binary vectors were clustered based on their correlation (color bar, Pearson's correlation). Clusters of vectors sharing neurons beyond a surrogate-based threshold were identified (colored squares, nonsignificant clusters were omitted). Each identified cluster of population event vectors represents a pattern, with the actual events being pattern occurrences. **c**–**h**, For each recording session, the number of pattern occurrences versus their within-cluster autocorrelation—reproducibility—was computed. Reproducibility of patterns in the two-photon (**c**–**e**) and Neuropixels (**f**–**h**) datasets from dorsal (VISp), medial (RS) and Fr (or ALM) areas for layer 2/3 pyramidal neurons (each panel reports the number of different cells, the number of experimental sessions and the median reproducibility value, dashed line). Fr, frontal cortical. FrMo, frontal motor cortical.

cortex, RSC), the Svoboda lab[32] (mouse anterolateral secondary motor cortex, ALM) and Neuropixels datasets collected simultaneously from all the mentioned areas and more by the CortexLab[26].

Reproducible population events throughout cortex

To identify population events and reproducible spiking patterns, we expanded a previously described method[7]. Briefly, population events were identified as peaks of synchronous firing that deviated significantly from a surrogate-based threshold (Fig. 1a,b; example from the MICrONS dataset). Neurons firing within an event were listed in an event vector (Fig. 1a, top spike raster, bottom firing rate). Each vector binarily encoded the state—active or inactive—of the individual neurons. We then used clustering by correlation, an unsupervised technique designed to group neuronal firing rasters without predefined categories (Fig. 1b). Finally, we selected clusters with a significant 'pattern reproducibility'—that is, with a within-cluster vector correlation beyond a surrogate-based threshold. Each cluster of population event vectors represented a pattern, with the actual events being occurrences of that pattern. A large portion of all recorded events, across both two-photon dataset (VISp = 0.45 ± 0.21, RSC = 0.64 ± 0.61, ALM = 0.70 ± 0.27) and Neuropixels dataset (means of V1/V2 = 0.42 ± 0.21, parietal = 0.37 ± 0.12, frontal = 0.48 ± 0.13), formed clusters with within-cluster correlation values (medians between 0.241 and 0.391; Fig. 1) exceeding those of surrogate controls. Structured recurrence is thus a generic feature of spontaneous cortical activity.

No pattern-completion units in cortex

The above results support the traditional hypothesis that cortical dynamics, with their reproducible patterns, are akin to attractor networks (we also confirmed such dynamics; Extended Data Fig. 1). Attractor networks can be tuned to display these dynamics, using biologically plausible Hebbian learning rules[9,19], to strengthen the connections among coactive units[8,9]. This process can create fixed point, limit cycle and continuous attractor dynamics in neural networks[11,18,33,34].

If reproducible cortical activity results from attractor dynamics, structural indicators should be found in cortical architecture. We first tested this assumption in the mouse visual cortex because it has provided connectivity evidence supporting the hypothesis of attractors in the cortex[27,35–37]. For this task, we used the MICrONS dataset[29]. This unique project precisely mapped three million synaptic connections using electron microscopy (EM) and coregistered the spiking activity of more than 100 neurons using two-photon calcium imaging.

At the cell level (Fig. 2), in line with previously reported statistics[27,35–37] (Fig. 2a,b), we found few connections among cotuned cells (Fig. 2c,d). At the population level, our analysis identified 226 significant events. These population events, numerous enough to yield statistical power (88%), showed significant specificity for orientation and direction[3,37] (Fig. 2e), with 27.1% cotuned cells (having orientation preference within 22.5° from the stimulus). Notably, the cells participating in

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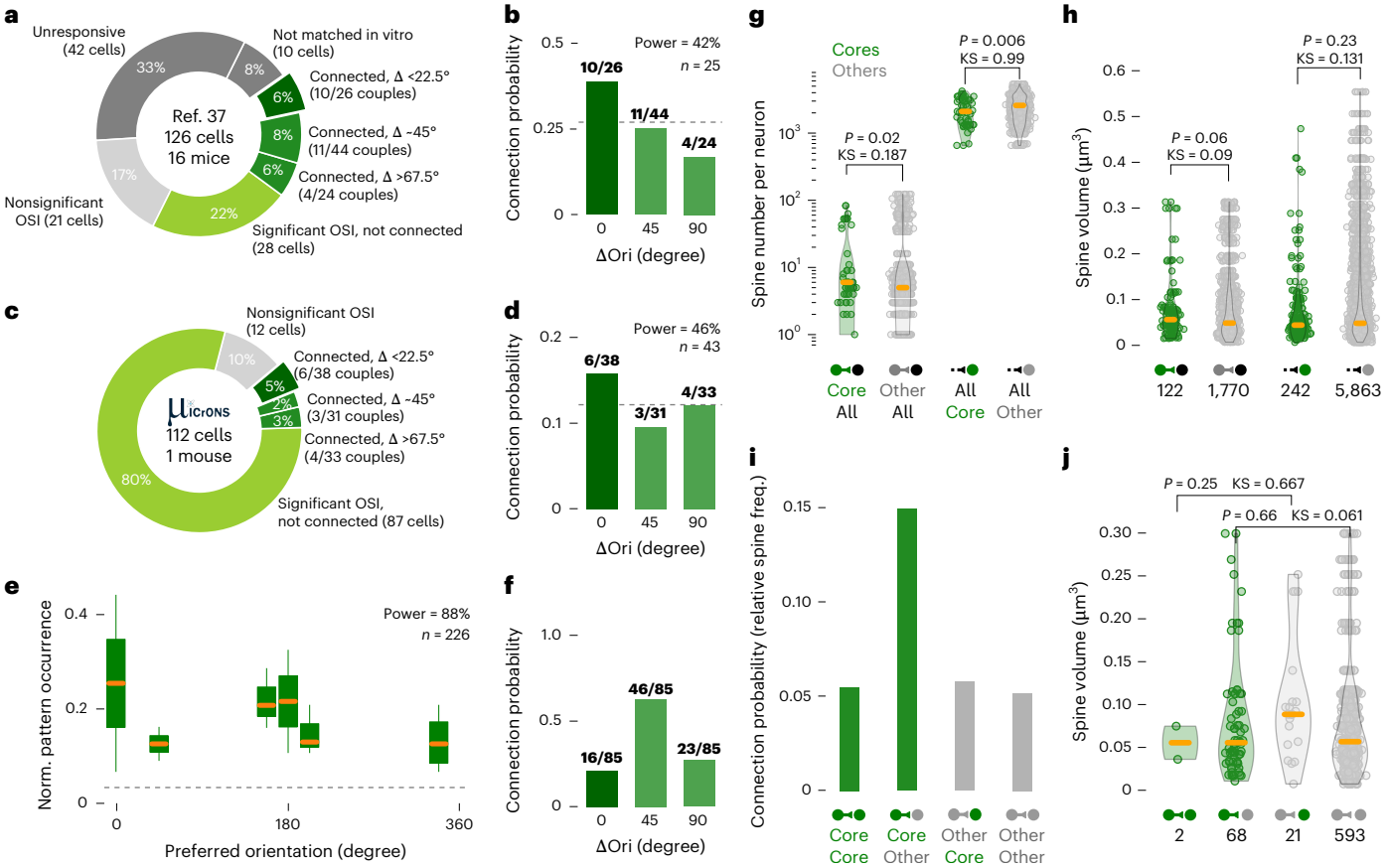

Fig. 2 | Cell connectivity does not support functional selectivity and pattern reproducibility. a, Categorization and number of cells sampled in the original study in ref. 36. Please note the number of cells presenting a significant OSI, the number of cotuned and connected cells (green notched slice) against the whole. **b**, Relationship between connection probability and difference in preferred orientation (ΔOri degree <22.5, from 22.5 to 67.5, and >67.5) between cell pairs responsive to grating stimuli (all pairs connection probability = 0.27, dotted gray line). The differences in orientation degree values are collected in bins of 45°). The sample for the reported trend is not statistically powerful (42%, *n* cells). **c**, Categorization and number of cells from the MICrONS dataset. Few cotuned and connected cells are present. **d**, The relationship between connection probability and difference in preferred orientation for the MICrONS dataset bears no trend and statistical power (46%, *n* cells). **e**, In the MICrONS dataset, the reproducible patterns (*n* = 226) were grouped around the horizontal axis (~0/360 and ~180°, statistical power 88%). **f**, Orientation preference of cells participating in orientation-selective reproducible patterns. The majority of participating cells had varying selectivity within 45° from the stimulus. **g**, Number of postsynaptic spines per neuron made by two-photon imaged cores (green) and other noncore (gray) neurons to all (black) neurons within the EM volume (left) and received from neurons within and beyond the volume (right). Cores received significantly fewer connections compared to other neurons. **h**, Postsynaptic (proofread) spine volume made by all cores and other neurons to all neurons within the EM volume (left) and received from neurons within the volume (right). There was no significant difference between spine volumes. **i**, Connection probability for core and other noncore neurons. **j**, Postsynaptic spine volume distributions of cores and others were not significantly different. For **g**, **h** and **j**, Kruskal–Wallis significance (*P* value) and KS effect size are reported.

these events demonstrated diverse orientation preferences, with 72.9% tuned around 45° away from the stimulus, highlighting the ability of population events to recruit cells based on more than individually cotuned responses (Fig. 2f). Thus, the dataset presented reproducible activities, usually taken as an indication of attractor dynamics.

We thus searched for indicators of attractor structure. Reproducible firing patterns do not repeat identically. Only a subset of their member cells—called 'core' neurons—reliably takes part in many occurrences of the same firing pattern[3]. Given this reliable participation in functionally specific events, core neurons are optimal candidates to serve as pattern-completion units[6,8,37]. We singled out core neurons as those participating in most events of significant clusters, and we confirmed core identities using functional correlation[38] (Extended Data Fig. 2). The MICrONS two-photon recordings presented 14 clusters of significant pattern reproducibility (>0.28), with 30 core neurons reliably occurring across events.

Once core neurons were identified, we proceeded with testing the following two fundamental attractor-driven assumptions about cortical connectivity: (1) 'synapses between cores are more numerous and/or stronger compared to others' and (2) 'cores have more recurrent connections'[8,36,39].

To evaluate the first assumption, we used the number and volume of postsynaptic spines[40,41]. When all EM-proofread synapses were taken under consideration, we had evidence (statistical power of 92%) to rule out any connectivity preference for core neurons (Fig. 2g and h). Notably, for following considerations, cores had a higher probability of having connections to other neurons than to cores (Fig. 2i), only two core neurons presented connections among themselves, and the postsynaptic spine volumes made by core neurons were not bigger than those made by other neurons (Fig. 2j).

To evaluate the second assumption, about the recursivity of circuits comprising cores, we performed several graph theory analyses, considering cells as 'nodes' and synapses as 'edges' of the network[42]. To get a global picture of mutual connectivity, we looked at the assortativity *r*—the proportion of mutual edges in a directed graph. We found $r = -0.08$, characteristic of nonmutually connected networks. A detailed analysis of motif distribution[43], across all population events

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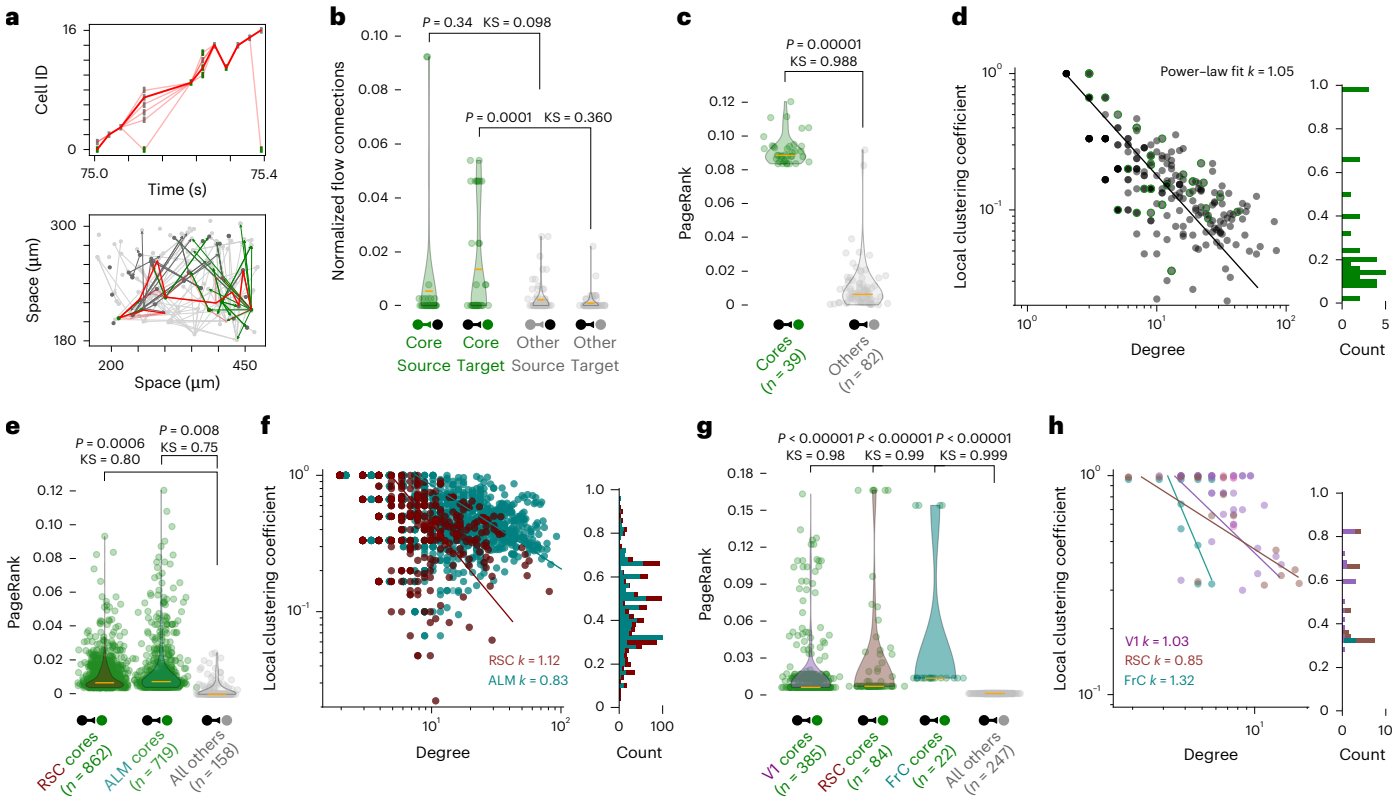

Fig. 3 | Reproducible patterns run through high-information flow nodes of modular networks. a, Population events can be studied as flows (red lines, several different flows are possible) in time (top)—from the first (bottom-left) to the last (top-right) firing cell—and in space (bottom) over the network. **b**, Neurons can be the source of outgoing and the target of incoming connections. By following all possible pathways between active cells in sequence, we retrieve each cell's flow capacity between source and target. The removal of connections targeting core neurons (green) disrupted the flow significantly more than that targeting other neurons (gray). Kruskal–Wallis significance (*P* value) and KS effect size are reported above the corresponding comparisons (in all following panels). **c**, The PageRank gives a weighted measure of the convergence toward cells participating in events. In the MICrONS dataset, core (green) neurons had significantly higher PageRank scores than others (gray). **d**, High-information flow nodes emerge from central positions in modular networks. Left, the log–log–linear relationship between the degree (number of connections) and local clustering coefficient (likelihood that neighbors of a specific node are themselves connected) is a measure of network modularity (reported power–law fit). Nodes with high degree and low local clustering coefficients (bottom-right) are found at the junctions between different modules in the network. Right, the local clustering coefficient distribution of core neurons is skewed toward low clustering values (skewness coefficient of 1.53). **e**–**h**, All selected datasets exhibited similar flow, PageRank and a log–linear relationship, with cores consistently displaying low clustering coefficients (two-photon datasets (**e**,**f**), Neuropixels datasets (**g**,**h**)). Same representations as in previous panels (*P* value reported as inequality below 1×10^{-5}).

of each cluster, including mutual, convergent and divergent motifs, showed no specific patterns for cores (Extended Data Fig. 3).

Despite lacking direct mutual connections, cores could have been sustained by indirect synaptic feedback, thereby creating cliques of pattern-completion units. These possibilities require the existence of more edges, paths and cycles that involve cores than noncores. We measured the number of cliques, the number and length of shortest paths and the number of cycles between cores and noncores. None resulted in a significant advantage to cores (Extended Data Fig. 4).

To summarize, we found that if we consider core—putative pattern completion—neurons, their functional and structural connectivity is not biased toward other cores nor cotuned neurons. These results challenge the applicability of the attractor model to the cortex. But a different approach is possible to explain the presence of population events.

Cores and the flow of population events

The large number of connections from cores to other neurons (Fig. 2i) suggested their potential centrality in cortical pathways. However, traditional centrality measures (degree, betweenness and hub scores[42]) showed no significant difference between cores and other neurons (Extended Data Fig. 5). But these static measures ignore dynamics. We thus analyzed cortical connectivity as a function of the temporal structure of population events. Events are sequences of cell activations in time and space (Fig. 3a, top and bottom, respectively). In the MICrONS dataset, we explored whether core neurons influence the flow of activity in the network. The amount of activity that can traverse edges between nodes is measured as 'flow'[44–48]. We measured the minimal flow cut, which is the minimal number of edges that must be removed to disconnect the subnetwork defined by each event, going from the first—'source'—firing neuron in the event subnetwork, to the last—'target'—firing neuron. We found that the removal of core connections disrupted the flow between cores as sources and all targets significantly more than other connections (Fig. 3b; $P = 0.0001$, Kolmogorov–Smirnov (KS) = 0.360). This result was confirmed by measuring the PageRank[43] of the neurons participating in each population event. This algorithm ranks the importance of each node by adding up the importance of every node that links to it, divided by the number of links emanating from it. Thus, PageRank represents a weighted measure of convergence within the neurons in an event. Core neurons had significantly higher PageRank scores than others (Fig. 3c; $P = 0.00001$, KS = 0.988).

We found that core neurons are not only important for information flow but also critical for maintaining the network's connectivity. Cores with high PageRank are typically convergence nodes in the network and central to information flow. A high minimum cut indicates that

removing these nodes would compromise the network's structural integrity. Together, these measures suggested that convergent connections could funnel incoming activities toward cores. We thus proceeded to better describe the networks embedding core neurons.

Cores at the gates of hierarchical modules

High-information-flow nodes typically emerge from controlling positions within modular networks. Nodes within modules have dense interconnections but low flow values. In contrast, intermodule nodes have high-flow values due to converging and diverging pathways. A measure of network modularity[38,49] is the relationship between the number of connections—degree (d)—and local connectivity arrangement—clustering coefficient ($C(d)$), the likelihood that neighbors of a specific node are themselves connected. As modules vary in size and interconnectivity, a power–law relationship between clustering coefficient and degree—$C(d) \sim d^{-k}$, with $k \sim 1$—points to hierarchical modularity. In the MICrONS VISp network, this relationship was $k = 1.05$ (Fig. 3d, left).

Once VISp modularity was established, we wanted to understand the role of core neurons in it. In the MICrONS dataset, where both activity and structure are known, core nodes with high flow exhibited high degree and low local clustering coefficients (Fig. 3d, right), suggesting their role as connectors and gatekeepers among different modules in the network[46,48].

We confirmed these results outside the visual cortex. The other sensory, associative and motor area datasets analyzed exhibited similar flow, PageRank and log–linear relationship, with cores consistently displaying low clustering coefficients (Fig. 3e, $P = 0.0006$ and KS = 0.8, and Fig. 3g, $P = 0.00001$ and KS = 0.98). Despite the connectivity being based on the functional rather than on the structural data, we ensured the robustness of connectivity estimation by checking paired short-time (1–5 ms) correlations in the Neuropixels dataset (Extended Data Fig. 6) and confirming that randomly rewiring the MICrONS dataset led to a loss of modularity (Extended Data Fig. 7).

Thus, we found that all inspected cortical networks do not present the structural signature of attractor networks but (1) have characteristics of hierarchical modularity, (2) have core neurons placed at the interface of modules and (3) have high-information flow and PageRank values. To unify these observations, we propose that distance-dependent connectivity is the fundamental requirement for creating such dynamics.

Modeling transient reproducible patterns

We offer a mechanistic understanding of how common neural network properties can unify the available evidence to yield transience and reproducibility.

There is an abundance of dissipative models with local connectivity from physics that can inform our understanding of neuronal dynamics in neuroscience[50]. However, these models leave out cortical population events transience and reproducibility, which are usually explained independently using ad-hoc connectivity[51], dynamically modifying synaptic plasticity[52] or with additional currents[53]. Based on the strong connectivity assumption of attractor networks, models have been proposed where strong recurrent paths were imposed on randomly connected networks by either additional circuits or increased synaptic strength between arbitrarily selected neurons[6,18,54]. These models yield synchronous firing events corresponding to active subpopulation(s). However, the explanatory and predictive power of these models is diminished by their ad-hoc solutions[25], producing only stereotyped stimulus responses, supporting a narrow set of input–output computations, and requiring strong inputs to exit the attractor state[55,56].

Here we present a chain of mechanistic hypotheses—distance-dependent connectivity, decaying in probability with distance, generates hierarchically modular networks where each module has multiple converging paths funneling toward high-information-flow units, which contribute to the reproduction of firing patterns by constraining the number of possible trajectories. We wove this hypothesis by threading known facts. Distance-dependent connectivity creates modules by allowing predominantly short-range, and few long-range, connections[57,58]. Such modular structure has been shown to occur among the cells of mouse[38], monkey and human[59] cortices. Crucially, in hierarchically modular networks, each module forms a convergent-divergent connectivity motif[49,60,61], arranged to funnel long and short-range activities toward the input/output nodes, characterized by high-flow values[48]. In a spiking network, these neurons would become more frequently active, leading to reproducible events.

We tested our hypothesis in spiking models by systematically varying the range of connection distances among neurons. To perform the tests, we developed a biologically informed, yet minimal, model of a cortical network of adaptive exponential integrate-and-fire model neurons[62]. In building our model, we established connections for each neuron by randomly sampling its neighborhood with probability exponentially decreasing for increasing distances (Fig. 4a). We chose nine connectivity ranges evenly spaced between 22 and 90 µm, after the EM distribution of connection distances (Supplementary Table 2 and Extended Data Fig. 8).

Across all ranges, we tested for the presence of population events and cores using flow statistics comparable to experimental results. Over the nine ranges, only four ranges produced synchronous irregular population events. Only to these four did we apply the same core analysis mentioned above. For example, the model with connectivity range of 50 µm (Fig. 4b) resulted in 431 significant events. Of these events, 273 made eight clusters of substantial reproducibility (>0.2), with 1,988 core neurons with population event duration distribution, cluster reproducibility, cluster coefficient and PageRank as in the MICrONS dataset (Fig. 4c and below). When exposed to the same stimulus twice, this network produced the same reproducible patterns (Extended Data Fig. 9).

Distance-dependent connectivity creates modules

Within the biologically realistic limits just described, we checked that connection distance is sufficient and necessary to produce modular networks (Fig. 4c). We found that all connectivity ranges produced hierarchically modular networks, with exponent k close to 1 ($k = (1.02, 1.3)$), as found experimentally[32] (Fig. 4d, black inset for the MICrONS data, $k = 1.05$). Distance-dependent connectivity was also necessary for hierarchical modularity. Beyond the physiological limit (>200 µm), modularity disappeared from the model due to the increased reach of connections and the consequent lack of local clustering, and a version of the same network with randomized connectivity preserving node degree presented no modularity (Supplementary Fig. 2).

Distance-dependent connectivity creates high flow

For the ranges that yielded hierarchically modular networks, with power–law relationships between clustering coefficient and degree similar to those observed in the MICrONS data, we inspected the connectivity organization within each module. As for the experimental data, model cores had a high degree and a low local clustering coefficient (Fig. 4e), indicating their location in the input/output portions of modules. Model cores were significantly more of the sources and targets of flow-cutting edges (Supplementary Fig. 1), and they had significantly higher PageRank (Fig. 4e), matching the structural-dynamical measurements of real core neurons (Supplementary Table 3).

The distance-dependent connection strategy was also necessary. We took the above network and randomly rewired all connections, preserving each neuron's node degree. This randomly rewired network gave rise to population events, but no reproducible pattern was identified, and consequently no cores (Extended Data Fig. 10). The reason is that by increasing the connectivity range, the number

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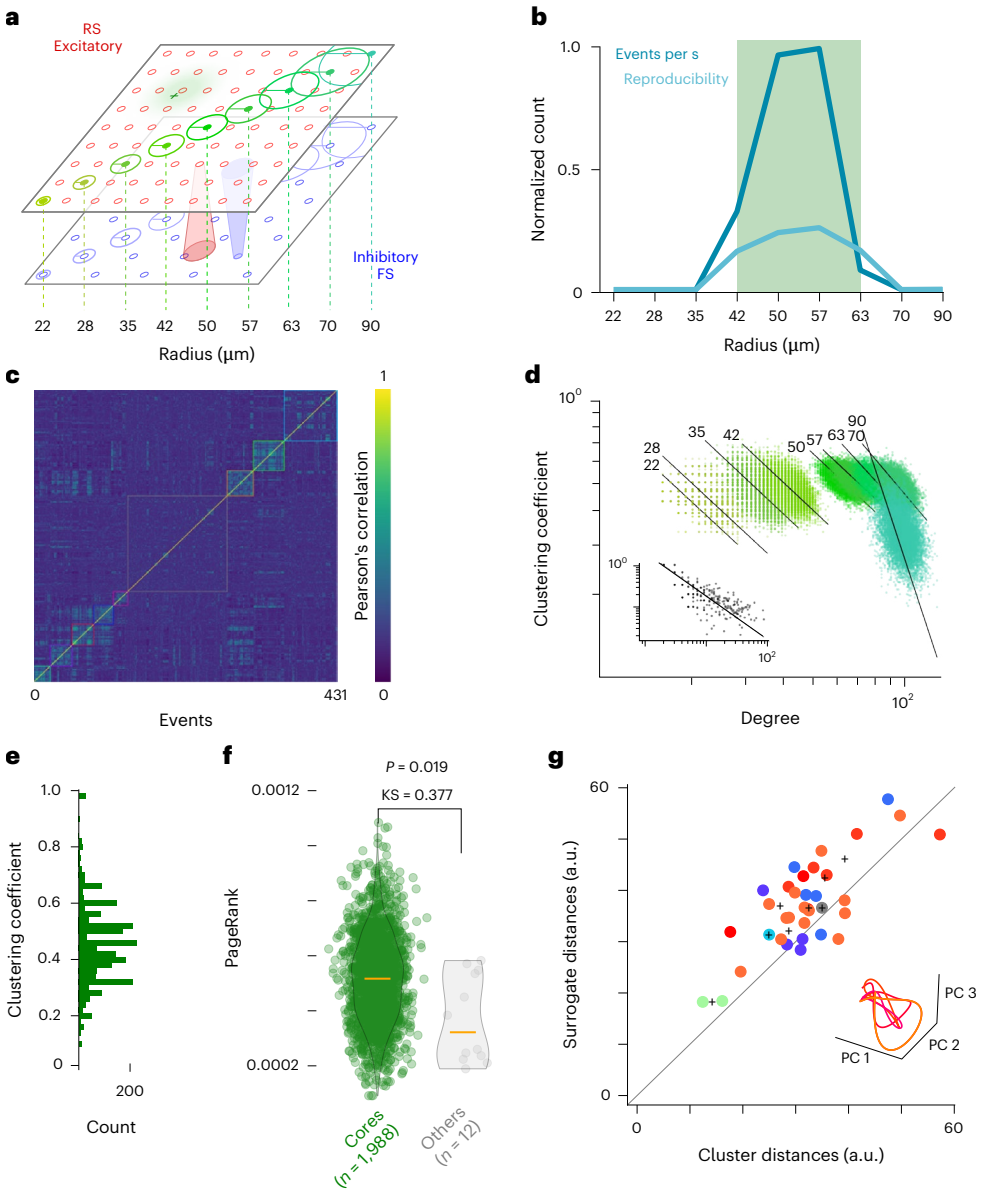

Fig. 4 | Distance-dependent connectivity creates high-flow core units. a, The models were made of excitatory (red) and inhibitory (blue) neurons on a 2D grid (100 × 100 μm), with a 10-μm step between neurons (10,000 neurons in total). Within biological limits, nine connection range radii were tested. **b**, Summary normalized number of events (dark blue) and reproducibility scores (light blue) peaked in the same connection ranges (42–63 μm) as the MICrONS data. **c**, Population binary vectors were clustered based on their correlation (color bar). Clusters of vectors sharing neurons beyond a surrogate-based threshold were identified (colored squares). **d**, All connection ranges resulted in hierarchically modular networks—that is, having a power–law relationship between degree and local clustering coefficient of their neurons—(inset, MICrONS data). **e**, The local clustering coefficient distribution of core neurons is not skewed toward low clustering values (skewness coefficient of 0.86) but is normally distributed toward low values (mean = 0.41). **f**, Core neurons had significantly higher PageRank scores than others (Kruskal–Wallis significance and KS effect size are reported). **g**, The set of simulated population event trajectories (examples in the inset) were more similar to one another than surrogate trajectories (**d**). The multidimensional (Hausdorff) vector distance between population event trajectories and that of the surrogate. 2D, two-dimensional; FS, fast spiking; RS, regular spiking.

of connections and the number of paths increase such that the funneling disappears[57,58]. Trying to rescue event reproducibility in random networks, for example, by increasing input correlations, led to population-wide oscillations far from biologically realistic regimes (Extended Data Fig. 10d).

Discussion

We examined transient and reproducible population events in local regions of the cortex. We found that these cortical areas exhibit hierarchical modularity and convergent information flows. Network models indicated that distance-dependent connectivity is necessary and sufficient to produce hierarchical modularity and transient population events with nodes of convergent information flow.

The substantial uniformity of cortical structure and function[63–65] similar findings in other brain regions[66], the genetic encodability of distance-dependent connectivity[67] and the early establishment of sensorimotor functional maps during embryonic development[68] allow us to speculate about the generality of these properties and identify a broader implication of our work. All animals need coordinated sensorimotor activity, a function often attributed to postnatal cortical learning[1,69]. However, the evidence also indicates that elements of such coordination are present at or even before birth, suggesting the

existence of preconfigured circuit motifs. We therefore propose a shift in perspective—population events may not only result from learning but may also act as its developmental substrate. These recurring events could scaffold the emergence of sensorimotor behavior by providing a structured initial dynamic, later refined through experience-dependent plasticity[51,70,71], allowing postaction evaluation mechanisms to shape synapses[72]. During life, this repertoire of patterns awaits future experiences. More specifically, learning may not construct the reproducible population events themselves, if these are prewired, but instead solidifies the transitions between them[51,73]. If cortex is prewired for recurrent co-activation to enable sensorimotor activities, our models suggest that distance-dependent connectivity allows the cortex to provide coordinated facilitation without prior learning.

Two objections might be raised about the results presented above. The first is that the available joint dynamical and structural dataset is limited to the visual cortex; other cortical regions show stronger mutual connectivity among functionally coactive neurons. However, our findings are consistent with previous studies that fail to detect strong mutual connectivity in cortex, and the visual cortex is often taken as an example of an attractor network for stimulus classification[74,75]. The second objection is that mutual connections between functionally correlated neurons have been observed; therefore, the point attractor model is valid. However, our analysis of the MICrONS dataset[36] (Methods) indicates that such connections are statistically too sparse to form the dense reciprocal motifs required by classical attractor models.

Our study faces several limitations. First, while two-photon imaging experiments can localize neurons and infer proximity, Neuropixels recordings lack the spatial resolution to assess intersomatic distances. Consequently, we cannot exclude the possibility that some correlated units may in fact be anatomically close and mutually connected. This spatial uncertainty limits our ability to generalize the anatomical conclusions from Neuropixels-based functional data. Future multimodal recordings combining fine-scale spatial resolution with high temporal precision will be essential to fully resolve this issue. Second, reliance on a single two-photon/EM mouse dataset may yield a limited and biased view, and third, the two-photon temporal resolution and EM sample size could be improved. Two series of facts mitigate these limitations. The substantial uniformity of measures across cortices and techniques supports the current results. In addition, our model, based on a few conservative assumptions, closely mimics experimental features both structurally and dynamically, and the distance-dependent strategy, adaptable to various cortical areas, invites exploration beyond the exponential sampling distribution (Supplementary Fig. 2). Finally, in another direction, synaptic connectivity is dynamic, so static EM data may miss transient connections—core neurons might be functionally connected during events, but such connections could be undetectable without time-resolved sampling. This would require an experimentally difficult strategy to obtain structural samples from the same tissue at widely different time scales. However, at the two time scales we considered, two-photon and electrophysiology datasets, the cortical networks produced reproducible activity without any sign of strong connections among the cores. Relatedly (5), it is possible that attractor dynamics are slower/faster than the resolution of the available recordings. We used two-photon and Neuropixels datasets—relatively slow and fast, respectively—to address this point, finding convergent results.

In conclusion, we provided evidence for high-information flows and hierarchical modularity in cortex. These results suggest how the brain's capacity for coordinated activity and learning is preconfigured by simple intrinsic network structures.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41593-025-02128-5.

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Methods

We performed the same structural and dynamical analyses over data collected in mouse cortex and in biologically constrained conductance-based spiking models.

MICrONS data source

The MICrONS project (https://www.microns-explorer.org/) made available through its website, two datasets, corresponding to the first two phases of the project. The first phase, the one we used, included EM and functional imaging of the same cortical patch of V1, in a 400 × 400 × 200 µm volume, with the superficial surface of the volume at the border of L1 and L2/L3, approximately 100 µm below the pia. Here follows a brief description of the procedures[30].

Two-photon imaging. Functional imaging was performed in a transgenic mouse expressing fluorescence GCaMP6f from the Jackson Laboratories (jax.org) and the Allen Brain Institute ('Allen Brain Institute data source'). The complete description of mouse transgenic lines, surgical procedures, implants, two-photon microscope features, stimulation and recording setups and protocols has already been published[30]. One important feature and the main difference with the Allen Brain Observatory two-photon dataset is the recording acquisition frame rate, here at ~14.8 Hz, resulting in time frames of ~67 ms (the Allen dataset has double this resolution, ~30 Hz). The mouse was head-restrained but could walk on a treadmill during imaging.

Stimuli. A total of 30 1-min trials of a colored noise with periods of coherent motion of oriented noise. Each 1-min trial contained 16 stationary-moving-stationary blocks, with a different direction presented in each block, pseudo-randomly ordered. Each of the 16 orientations was presented during 15 frames, followed by 40–41 frames of pink noise. All orientations were randomly repeated 30 times. All scans were performed using the same 16 orientations.

EM. After functional imaging, the mouse was perfused with fixing agents, dissected and 200-µm coronal slices of the V1 area, where the two-photon sessions were performed, were taken. Serial sections of 40 nm thickness were cut from the slices and prepared for transmission EM. Custom modifications to the setup and camera allowed fields of view as large as (13 µm^2) at 3.58 nm resolution. Data were acquired in blocks, aligned through multiple steps, and defects were manually detected. Somas, synaptic clefts, and spine detection and partnering were manually performed on a subset of 387 clefts for training and testing machine learning libraries. The rest were performed by machine learning libraries. Finally, cells identified in the two-photon imaging max projection were manually coregistered with EM-identified somas.

Allen Brain Institute data source

The Allen Institute made available, through its Brain Observatory, a large dataset of neuron responses from several visual cortical areas and layers, using mouse transgenic lines expressing a Cre/Tet-dependent fluorescence calcium indicator (GCaMP6f), which allows the recording of calcium influx associated with neural activity. The complete description of mouse transgenic lines, surgical procedures, behavior training, implants, two-photon microscope features, stimulation and recording setups, and protocols were already published (see the online documentation at http://observatory.brain-map.org). Here we briefly provide the relevant information in the context of our work.

Cell type and experiment selection. The Allen Brain Observatory contains data collected in several cell types of mouse visual cortices (Allen VisualCoding_Genotyping whitepaper). Cell-type-specific expression of GCaMP6f was conferred by the Cre-recombinase and tetracycline-controlled transactivator protein under the control of the Camk2a promoter with the Ai93 and Ai94 reporters. To maximize the number of recorded neurons across layers, we selected all Cre drivers specific for pyramidal neurons and layers. Considering that each layer of mouse V1 is roughly 100 µm thick[76], we were able to design cell-type-specific and layer-specific queries that resulted in the selection of the following experiments (as of 9 October 2019). We selected layer 2/3 planes to have a dataset comparable to the one from the MICrONS project (http://observatory.brain-map.org/visualcoding/search/overview?area=VISp&imaging_depth=175,185,195,200,205,225,250,265,275,276,285,300&tld1_name=Slc17a7-IRES2-Cre,Cux2-CreERT2,Emx1-IRES-Cre). The selected experiments are 502115959, 501574836, 502205092, 502608215, 510514474, 503109347, 510517131, 524691284, 645413759, 660513003, 704298735, 702934964, 712178511, 526504941, 528402271, 540684467, 545446482, 561312435, 596584192, 647155122, 652094901, 652842572, 653122667, 653125130, 657080632, 661328410, 661437140, 663485329, 679702884, 680156911, 683257169 and 688678766 (of note, the experiment 501271265 was excluded due to the issue reported in https://github.com/AllenInstitute/AllenSDK/issues/66). A total of 32 experiments were selected and downloaded as NWB files. Each experiment contained preprocessed calcium indicator fluorescence movies (512 × 512 pixels, that is, 400-µm cortical field of view, sampled at 30 Hz), which were corrected for pixel leakage during scanning, filtered based on regions of interests and had the neuropil signal subtracted. Each resulting NWB file consisted of an experiment session (~90 min) at a given location (cortical area and depth).

All original codes of this study have been deposited at https://github.com/dguarino/Guarino-Filipchuk-Destexhe and are publicly available as of the date of publication. DOIs are listed in the key resources table. The MICrONS project dataset is available at https://www.microns-explorer.org/phase1. The Allen Brain Observatory dataset is available at http://observatory.brain-map.org. The calls to retrieve the Allen data are listed in the Methods. An interactive Jupyter notebook is accessible on CodeOcean (https://codeocean.com/capsule/9782876/tree). The code to run all simulations (complete with Dockerfile to recreate the environment) is available at https://github.com/dguarino/Guarino-Filipchuk-Destexhe-simulations. Any additional information required to reanalyze the data reported in this paper is available from the lead contact on request.

Other data sources

To broaden our analysis to other cortical areas and ensure the replicability of our results, we used two additional sources from Svoboda's and Goard's labs. For the ALM cortex, we used the dataset analyzed in ref. 33, available for download on the CRCNS website (https://portal.nersc.gov/project/crcns/download/alm-2). For the RSC, we used the dataset analyzed in ref. 32, available for download on the Dryad data repository (https://datadryad.org/stash/dataset/doi:10.25349/D99W4T). For the Neuropixels recordings, we used the data from ref. 27, available for download on the FigShare data repository (https://doi.org/10.25378/janelia.7739750.v4). Given our interest in firing patterns, we used the available preprocessed unit firing, in which waveforms were used to identify units.

Calcium spike timing extraction

To infer the action potential timings from the calcium indicator fluorescence of the Allen Brain data source, we used the Python library OASIS. This library for spike inference, based on an active-set method, also includes routines for automatically estimating its parameters. The library is available at https://github.com/j-friedrich/OASIS. As parameters for the deconvolve function, we used penalty = 1, and we used the automatic estimation of fluorescence variation from the data (parameter g). The resulting spike timings obtained with these parameters were verified against the example deconvoluted data provided in the Allen Brain Observatory IPython notebooks.

Dynamical analysis

For each data source, we applied the same dynamical analysis. The analysis of synchronous irregular population events has already been developed in MATLAB and discussed in ref. 7. We reimplemented all analysis steps in Python and tested them against the original analysis. Briefly, single-cell activity was described as a cell binary vector, where the number of elements was equal to T, the number of time frames during the recording (frame duration was ~30 ms). A value of 1 was assigned at the time frames corresponding to the onset of each spike, and the cell vector was zero otherwise. The number of cell vectors was equal to the total number of recorded neurons in the field of view, n_e, resulting in a $T \times n_e$ spike train matrix, where columns corresponded to the time frames and rows to the neurons. Each column was summed to yield the number of coactive neurons at each time frame, from which the instantaneous population firing rate could be deduced. As shown in sample raster plots (Fig. 1a), there were periods when spikes were much more synchronized across the population, which could be interpreted as population events, distinct from the fluctuations of an asynchronous population spiking process. To identify whether the population events were above chance coincidence and determine their exact start and end, we applied the following algorithm: (1) we created 100 surrogate $T \times n_e$ spike trains matrices by reshuffling each neuron's interspike interval. The average 99 percentiles (several other percentiles can be tested in the available IPython Jupyter notebooks) across time frames were then calculated and a firing rate threshold for event significance was obtained by adding this average value to a local baseline estimate that aimed at correcting slow fluctuations of background firing rate. The baseline estimate was generated using an asymmetric least square smoothing algorithm with parameters $P = 0.01$ for asymmetry and $\lambda = 10^8$ for smoothness. (2) Experimental population firing rate trace was smoothed using Savitzky–Golay filtering, with order = 3 and frame window = 7, to get rid of nonessential peaks. Local maxima for the smoothed curve that were above the previously defined threshold were retained (Fig. 1b, red square). The adjacent local minima around each of the local maxima were identified (Fig. 1a). Their time frames were taken as the beginnings and ends of population events (Fig. 1a, limits of the green vertical band). (3) The time interval between the beginning and end of each population event defined its duration. We described each population event as a binary vector of length equal to the total number of neurons in the recorded field of view (n_e). We initialized all vector elements to 0, and then we indicated the participation of a cell in the event, spiking at least once over its duration, with a coordinated 1 (Fig. 1a). (4) Pearson's correlation matrix for ongoing and evoked events was constructed by computing the Pearson correlation coefficient among all binary vectors. In plotting the matrix, the color reflected the degree of correlation with a colourmap ranging from 0 to 1. (5) Population events sharing a substantial number of neurons, that is, having a significant degree of correlation, were clustered together (using agglomerative linkage clustering with the farthest distance based on the correlation metric, from the Python 'scipy.clustering' package), and the correlation matrix could be permuted according to the clustering order. (6) The correlation between population events might be simply due to the finite number of recorded cells. We, therefore, defined a significance threshold for intracluster correlation, established using surrogate events. We generated $100 \times n_e$ surrogate event vectors (with the same number of cells per event of the originals, but randomly chosen cells, without replacement). We then performed correlation-based clustering (as in point 5) of the resulting surrogate clusters, and we used the 95 percentiles of the correlation coefficient of these clusters as a correlation threshold (other values in the available IPython Jupyter notebooks). For each population i, we extracted the following three figures: number of events (events_i), total number of clusters (clusters_i), and mean cluster size (mean number of events per cluster, clustersize_i), and then computed the $\text{ratio}_i = \frac{\text{clusters}_i \times \text{clustersize}_i}{\text{events}_i}$. (7) To quantify the pattern reproducibility of events within a cluster, that is, the number of cells shared across events in a cluster, the correlation was calculated across all events of each cluster. (8) Core cells were identified as those participating in a range of thresholds (from 55%, barely above no difference in participation, to 95% in the available IPython Jupyter notebooks) of the events within each cluster. Relaxing this requirement to allow for a more participatory definition of cores changed the statistics but not the trends presented in Fig. 1. Once identified, we verified that the average firing rate of each core was higher within its events than outside, discarding those with unspecified firing. In the MICrONS dataset, there are five two-photon scans, corresponding to different depths. In the first scan, we found 3.86 ± 1.96 (min 0, max 6) cores per cluster, 31.14 ± 1.96 (min 29, max 35) others per cluster. In the second and third scans (grouped to preserve statistical significance), cores per cluster = 1.00 ± 1.41 (min 0, max 3) and others per cluster = 20.00 ± 1.41 (min 18, max 21). In the fourth scan, cores per cluster = 1.50 ± 0.50 (min 1, max 2) and others per cluster = 20.50 ± 0.50 (min 20, max 21). In the fifth scan, cores per cluster = 0.75 ± 0.83 (min 0, max 2) and others per cluster = 11.25 ± 0.83 (min 10, max 12). In developing our dynamical analysis for the MICrONS data, we used the available notebook example (github.com/AllenInstitute/MicronsBinder/blob/master/notebooks/vignette_analysis/function/structure_function_analysis.ipynb). The whole analysis workflow code is available on CodeOcean (codeocean.com/capsule/9782876), which can take a considerable amount of time to download the data from the online resources.

Attractor analyses

In experimental data, where no system's characteristic equation is given, we used two levels of analysis. For a network of size n, we defined a 'state' vector composed of the activity of n units. Each 'state' comes from a two-photon frame, deconvolved into continuous dF/F values for the n neurons.

(1) At the 'population event level', we study the cell signature of an event. An event e is a set of states. At the population level, a vector is formed as the average of all set states. To determine whether multiple events form clusters, here we projected each event into a reduced set of embedding dimensions identified using principal component analysis based on the population vectors. We then checked whether the clustered events above occupy different portions of the reduced-dimensional space (and were colored accordingly). At this level, an attractor is the clustering of events with an average Silhouette score of more than 0.7 (for additional details on this method, see the companion Jupyter notebooks, codeocean.com/capsule/9782876).

(2) At the 'trajectory level', we studied the approach to and the dynamics of each event (as in ref. 77). Here we studied whether sequences of frames belonging to events from the same cluster have similar trajectories. To quantify the similarity among event trajectories, we adopted the same strategy discussed in ref. 78. If noise is the source of discrepancies in the frame trajectories of events from the same cluster, then their distances in the embedding space should be equal to those between artificially recreated trajectories. We measured the degree of overlap between their trajectories in the reduced-dimensional space of principal component analysis using the Hausdorff distance. To establish the significance of these distances, we compared them to the distances among surrogate trajectories obtained by shuffling (10,000 times) the original frame distances. We established confidence intervals for the Hausdorff distances between actual and surrogate dynamical trajectories using bootstrapping. Colored dots indicate distances between pairs of frames in the set of events from the same cluster; color indicates cluster.

Saddle node analysis

First, we computed the time 'derivatives' of the neuronal activity data. This was achieved using the central difference method. The derivative for each time point t was estimated as $f'(t) \approx \frac{f(t+h)-f(t-h)}{2h}$, where h represents the uniform time step size. For boundary conditions, the first and last derivative values were computed directly, while intermediate derivatives were calculated as the average of adjacent central differences. Then, we computed the Jacobian matrix of the system, which represents the partial derivatives of neuronal activity dynamics. For each time point and pair of neurons (i,j), the partial derivative was computed as: $\frac{\partial \dot{x}_i}{\partial x_j} = \frac{\dot{x}_i(t+1)-\dot{x}_i(t-1)}{\dot{x}_j(t+1)-\dot{x}_j(t-1)}$, where $\dot{x}_i$ represents the derivative of the activity for neuron i, estimated using the $\Delta F/F$ from the two-photon data. Finally, we examined the stability of the neuronal dynamics using the computed Jacobians. For each Jacobian matrix, the eigenvalues and eigenvectors were computed using the NumPy 'eig' function. The eigenvalues were analyzed to identify positive real parts (indicative of instability) and negative real parts (indicative of stability). Time frames exhibiting both positive and negative eigenvalues were classified as saddle node states. These frames were noted for further analysis. For each detected saddle node state, the contributions of individual neurons were assessed based on the eigenvectors associated with positive eigenvalues. Neurons contributing more than 5% to the eigenvector magnitude were identified. Saddle node states were analyzed for overlap with predefined event time windows. A permutation test (1,000 iterations) was conducted to assess the expected overlap of saddle node states with events by chance. The proportion of saddle node-contributing neurons overlapping with known cluster cores was calculated for each scan. Observed and expected ratios of saddle node states within events were computed, and statistical significance was assessed using permutation results. Only data from the MICrONS project were used, where the $\Delta F/F$ data were made available, while the other datasets only provided processed spike trains.

Orientation tuning (ORT) and cell pairs connection analysis

For the MICrONS dataset, we computed for all 112 two-photon recorded VISp cells, the ORT[79]: $\mathrm{ORT} = \frac{1}{T}\sum_t^T R_{td}$, where T is the number of trials, R_{td} is the number of spikes fired during one trial presentation of one stimulus with orientation d, and the orientation sensitivity index (OSI)[80]: $\mathrm{OSI} = |\frac{\sum_d^D R_d \times \exp(i\frac{\pi}{4}d)}{\sum_d^D R_d}|$, where R_i is the response during stimulus presentation d. To test OSI significance, we computed 100 (or 1,000) permuted spike trains. We then performed the same connectivity analysis as in ref. 36. Connection probabilities were calculated as the number of connections detected over the number of potential connections attempted. Having access to all cells simultaneously, we sorted all orientation-tuned cells by their OSI. Then, we counted the unidirectional/bidirectional connections between all combinations of cells. The connection probability was computed over the total number of couples. Using the computed probabilities, we tested the statistical independence of the connection probability among cells with different orientations (using the chi-squared test for statistical independence). The ORT of population events was computed similarly. See the companion IPython notebook for details on the computations (codeocean.com/capsule/9782876).

Statistical analysis: general

For all statistical tests, to evaluate the differences among distributions resulting from our measurements of the nominal variables (spine volumes, 1-lag correlations, degrees, betweenness, ...) we considered that the nature of the outcome was continuous, the number of groups was larger than 2 (unless specified, or with unequal measurements) and the groups were independent (cores and noncores groups do not share cells). We tested the distributions' normality using the D'Agostino and Pearson's test. For normally distributed variables, we used the one-way analysis of variance test. For non-normally distributed variables, we adopted the Kruskal–Wallis test. For some variables, the sample size was large (exceeding 100 samples); therefore, even small variations could be deemed significant by the above tests. In these cases, a quantitative measure of the effect size was required. As a measure of effect size, we adopted the KS test difference, which is more robust than Cohen's d for non-normal distributions[81]. For all these tests, we used the Python 'scipy' library.

Power analysis: cell pairs

We evaluated the results from ref. 36 from the statistical power point of view. From their article: 'The data set contained imaging experiments performed on 16 mice and whole-cell recordings from 126 L2/3 pyramidal cells, 116 of which could be matched to neurons functionally characterized in vivo (Methods). The rate of connectivity was 0.19 (43 connections out of 222 potential connections assayed), in keeping with previous reports[6,10]. Connection probability, synaptic strength and electrophysiological properties of OGB-1-labeled neurons were not significantly different to those recorded in slices from naive age-matched visual cortex that was not injected with OGB-1 AM (connectivity rate 0.18; 25 connected of 143 tested; Supplementary Fig. 2), indicating that dye loading, anesthesia and prolonged exposure to infrared laser light during imaging in vivo did not alter these parameters. We first examined how connectivity depended on orientation selectivity and on responsiveness to natural movies. Out of the 116 neurons, 77 were responsive to the natural movie, and 79 were orientation selective for grating stimuli (Methods). Connection probability between orientation-tuned neurons was more than twofold higher than among nonselective and/or nonresponsive cells (0.27; 25/94 versus 0.10; 3/31; $P = 0.050$, chi-squared test). The connectivity rate between neurons responsive to the natural movie was significantly higher than among cells nonresponsive to the movie (0.28; 30/108 versus 0.04; 2/48; $P = 0.001$, chi-squared test). Taken together, these data indicate that reliably responsive and feature-selective neurons belong to more densely interconnected neocortical subnetworks. We then related connection probability to neuronal preference for the angle and direction of drifting gratings (Fig. 2). For this analysis, we only included pairs in which both neurons were responsive (74/113), orientation selective (orientation selectivity index (OSI) > 0.4; 53/74), or direction selective (direction selectivity index (DSI) > 0.3; 41/53; Methods; Supplementary Fig. 3a–c)'. Therefore, a total of more than 16 mice and 126 cells were recorded. A total of 113 cells were recorded both in vivo and in vitro. A total of 74 cells were responsive, of which 53 were orientation selective. The actual number of cells found to be cotuned and connected is not said in the article. But we can compute an estimate, based on the information provided in Methods and Fig. 2a. It is said that 26 connection couples were tested for cotuned cells (and 44 and 24 couples for cells with increasing Δs). Considering that in the Methods it is said 'On average we patched 7.9 neurons (range = 2–14) and assayed 13.9 potential connections (range = 2–31) per slice' (corroborated by Fig. 2a and Supplementary Fig. 5), we can estimate that between seven and eight cells were found to be cotuned and connected (${}^8C_2 = 28$ and ${}^7C_2 = 21$), less than ten for Δ<67.5°, and about seven cells for Δ>67.5°. We wanted to understand whether the number of orientation selective, cotuned and connected cells found in the study, that is, 6%, or ~8 cells of 126, is sufficient to support the claim that local connectivity in the cortex is functionally specific. We took as explicit the usually tacit assumption that cortex, in general, and V1, in particular, is something like a pattern recognition network, or (fixed point) attractor network, much like the successful Hopfield model. We aimed to provide a simple estimate of the required number of samples to be performed in order to trust the trends found in the study. The goal was to achieve a desired level of statistical power (for example, 80%) while maintaining a chosen significance level (for example, $\alpha = 0.05$). The total number of samples will depend on how many orientation patterns were investigated. Because of the eight distinct orientation patterns discussed in ref. 36,

the sampling process needs to be repeated for each pattern. We had to make explicit assumptions—~5-µm thickness for a two-photon plane of 0.5 × 0.5 mm, giving roughly 10,000 neurons for the network[82]. We assumed a Hopfield-like network of 10,000 units with 5% sparse random connections. Given the 5% connectivity, each neuron is connected to roughly 5% of the other 9,999 neurons, which is about 500 neurons. We assumed that only a proportion of them have strong connections due to their similar orientation preferences (arbitrarily high 10% = 50 neurons). These percentages need to be applied to each of the eight orientation patterns. We can provide a simplified estimation of the required sample size for one orientation pattern with the given parameters assuming a standard normal distribution—(1) anticipated mean of connected units (μ_0) = 50, (2) s.d. (σ) = 0.025, (3) significance level (α) = 0.05, (4) desired statistical power ($1-\beta$) = 80% (Z = ~0.845) and (5) expected effect size ($\mu_0 - \mu_1$) = 0.5 (assuming a large effect size results in smaller samples). Our hypotheses, assuming a normal distribution of connections, were as follows: null hypothesis (H0)—the proportion of significant connections between coactive and cotuned neurons is equal to or lower than what would be expected by chance (for example, 5% due to sparse connectivity). Alternative hypothesis (H1)—the proportion of significant connections is higher than expected by chance. To estimate the required sample size, we used the Python power analysis library 'statsmodels'. We used the 'TTestPower' class to perform a power analysis for the one-sample t test. It calculates the required sample size based on the specified effect size, α level and desired power, assuming we are interested in comparing a sample to the hypothesized population mean (random connections). It resulted in a required sample size for each orientation—33, total sample size (eight orientations); 267, assuming an s.d. (Σ) of 0.086. A supplemental post hoc power analysis was conducted using 'G*Power' (version 3.1.9.6) to determine the actual power tested by the study. Results indicated that, with the sample size of n = 25 cells (circa 2 per orientation), at a significance criterion of α = 0.05, for a one-sample t test, the power was 42.5% (all these can be replicated in the online IPython Jupyter notebooks).

Power analysis: population events

We need to determine how many patterns are sufficient to claim that they are functionally linked to the presented oriented patterns. In this case, the effect size is related to the temporal dependencies between the firing patterns and the presented oriented patterns. We counted the number of events per stimulus orientation at increasing lags and normalized it by the total (to visualize the event counts for each orientation, see companion IPython notebook). We used the spread (number of lags with nonzero event counts per orientation, normalized by the total) after a stimulus presentation to quantify the effect size. To know if the occurrence of events meets the required power, we used surrogate data to set a threshold. With a sample size of 226 actual events, a significance of P = 0.05, and 10,000 random normal surrogate events (a normal distribution assumption, t test), we obtained a power (spread effect size), that is, 0.8896. Thus, the number of times an ensemble followed an oriented stimulus was sufficient to investigate the (functional) link between events and oriented stimuli. The specificity and significance of pattern occurrences (event occurrence as members of clusters) for oriented stimuli gave four significantly orientation-tuned patterns for 7 of 16 orientations (8 orientations for two directions[36]).

Statistical analysis: cores and spines

The data released during phase 1 by the MICrONS project concerned one mouse. This limit was our major concern while designing our study. However, our unit of statistical analysis was the number of events. Events are produced by the same cortical tissue overall, but by different cells and connections. This implies that we did not set out to measure the same outcome multiple times on the same animal—which would entail an assay variability. Instead, we set out to measure different (quasi)-independent outcomes, which have inherent biological variability (the high level of interconnectivity does not make it completely independent). This is why, in the currently released MICrONS dataset, we consider our sample size of n = 226 events to be statistically legitimate, that is, the number of independent observations of our unit of analysis under a single experimental condition. The goal of our study was to test hypotheses; therefore, we designed each analysis as a statistical hypothesis test to produce an exact statistical significance level (P value) with an appropriate effect size. Concerning the attractor features for cores versus noncores, they are taken to exist whether the following points hold in the specific direction (one-tailed direction): (1) core neurons have more numerous synapses between themselves, or toward others, compared to those of noncores. (2) Spines with presynaptic cores and postsynaptic cores or noncores are bigger than spines with presynaptic noncores. In Fig. 2g,h, we began by assessing whether core neurons made (or received) more numerous or larger spines, on average, compared to any other neuron (irrespective of whether they were part of the two-photon imaged dataset or not). The required sample size, common to both points above (but calculated here for the spine volume), can be determined by considering the following: the study group ('cores') is compared to the rest of the population ('all'). The primary endpoint is a continuous value (synaptic spine volume, µm^3). The anticipated mean (μ_0) of the 'others' is 0.05 (based on the recorded average, with σ = 0.25). Estimating a higher average for cores of 0.1 (μ_1), with a type I error probability (false-positive rate) of α = 0.05, a type II error probability (false-negative rate) of β = 0.25 (corresponding to a 75% power), the resulting sample size for core spines is n = 111, as $n = \frac{\sigma^2(z_{1-\beta}+z_{1-\alpha/2})^2}{(\mu_0-\mu_1)^2} = \frac{0.25^2(0.67+1.96)^2}{(0.05-0.1)^2}$. And the smallest count of spines (core-to-all; Fig. 2b) is n = 122. In Fig. 2i,j, we then examined the (core-to-core, core-to-other, other-to-core, other-to-other) connectivity of neurons participating in significantly reproducible events, hence limited to both EM and two-photon-imaged neurons. The required sample size, common to both points above, can be computed by considering the following. The study group ('cores') is compared to the rest of the population (noncores, or 'others'). The primary endpoint is a continuous value (synaptic spine volume, µm^3). The anticipated mean of the 'others' is 0.02 ± 0.12 (based on the recorded average). Estimating a higher average for cores of 0.08, with a type I error probability (false-positive rate) of α = 0.05, even with a type II error probability (false-negative rate) of β = 0.3 (corresponding to a modest power of 70%), the resulting sample size for cores is n = 254. The actual number of two-photon recorded cells is n = 112, but the number of cores per event varies between 3 and 8. Therefore, in the status of the MICrONS project, the spine count for cores is underpowered. To provide an estimate, we considered the following four aspects: (1) given that the spine count for each cluster of events with its cores and others is taken independently, many spines will be considered in a repeated measure fashion; (2) the maximum value of the independent variable (spine volume) is >10 times its minimal value (0.35 and 0.02 µm^3, respectively), allowing a large space of possible values; (3) we lowered the threshold for core identification to the minimum (participation to 60% of the events) and (4) we did not differentiate spines based on their point of contact (for example, axo-somatic, axo-dendritic, axo-axonic, etc).

Structural analysis

We used the data tables published by the MICrONS project on their website and the referenced publications (see above).

Postsynaptic spine number and volume analysis. We selected the subset of proofread synapses with soma root IDs that were identified and coregistered in the two-photon dataset. After the dynamical analysis was performed (see above), core neuron IDs for each event were available and used to query the synapse table, and postsynaptic spine volumes were collected by identified type (core or noncore). The four possible combinations (core–core, core–other, other–core,

other–other) were plotted in Fig. 2. Given that different cores participated in different clusters, the number of cores and others differed, and each core (or noncore) neuron could potentially be connected to all other cores (or noncores) we reported in Fig. 2 the normalized number of spines. Differences in significance and effect size were established using the Kruskal–Wallis test and the KS distance, as the distributions were non-normal, as reported below in the statistical methods.

Synaptic functional efficacy analysis. To estimate functional connectivity, we adopted the same method discussed in ref. 38. Briefly, an adjacency matrix representing a graph for each experiment was created with nodes being all recorded cells, and directed edges formed according to a single frame lagged correlation (frame duration, 67.4 ms) and weighted according to the number of spikes (Extended Data Fig. 2).

Graph theory analyses. An adjacency matrix was built by coordinating presoma and postsoma root IDs. All graph analyses were performed using the Python version of the library 'iGraph'[83]. The iGraph library allows motif identification with either three or four nodes. We used three motifs to facilitate comparison with other publications and the online MICrONS IPython notebooks, as well as to enable computations of the 100 surrogate graphs within a reasonable time frame. The comparison of core-based cycles and other-based cycles was performed over an adjacency list, using a function that recursively checked the list of neighbors starting with each core (or noncore) cell ID. In developing our structural analysis for the MICrONS data, we began with the notebook example on the Allen Institute MICrONS GitHub page (https://github.com/AllenInstitute/MicronsBinder/blob/master/notebooks/intro/MostSynapsesInAndOut.ipynb). The whole analysis workflow code is available on CodeOcean (codeocean.com/capsule/9782876).

Functional connectivity. To construct the functional adjacency matrix, we analyzed binary spike train data for pairwise correlations. The process involved the following steps: each pair of binary spike trains (i, j) was subjected to cross-correlation analysis (with maximum lag of 1 ms in the Neuropixels dataset, 30 ms in the two-photon dataset). Self-connections ($i = j$) were excluded by assigning a value of 0 for the diagonal elements of the matrix. The results of the pairwise cross-correlation analysis were stored in a two-dimensional array, forming the functional adjacency matrix. To ensure the sparsity of the adjacency matrix, weak correlations (<0.4) were set to 0. This threshold was based on the approach discussed in ref. 38.

InfoMap analysis. To identify structural modules in the MICrONS data and the models, which are directed networks, we used a random walk approach discussed in ref. 45. We also used the same algorithm to identify, within each module, the largest connected component and the in-component/out-component to the central one.

Network models

All simulated networks were composed of $n = 12{,}500$ (excitatory/inhibitory neuron ratio of 4:1, that is, 10,000 excitatory and 2,500 inhibitory) conductance-based adaptive exponential integrate-and-fire point neurons. The set of equations describing the membrane potential (1) and spike adaptation current (2) of a neuron are:

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -g_L (V - E_L) + g_L \Delta \exp((V - V_T)/\Delta) - w/S \quad (1)$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{1}{\tau_w} (a (V - E_L) - w) \quad (2)$$

with C_m being the membrane capacitance (nF), g_L the resting (leak) conductance (μ_S), E_L the resting (leak) potential (mV, which is also equal to the reset value after spike), Δ the steepness of the exponential approach to the threshold (mV), V_T the spike threshold and S membrane area (μm^2). When V reaches the threshold, a spike is emitted, and V is instantaneously reset and clamped to the reset value during a refractory period (ms). Furthermore, w is the adaptation variable, with time constant τ_w (ms), and the dynamics of adaptation are given by parameter a (in μ_S). At each spike, w is incremented by a value b (in nA), which regulates the strength of adaptation. An experimentally informed parameter set was chosen[84] and kept fixed, as shown in Supplementary Table 1.

Synapse model

The synaptic connections between neurons were modeled as transient changes in conductance. The synaptic time course was modeled as an α function with a fast rise followed by an exponential decay:

$$\tau_{\mathrm{syn}} \frac{\mathrm{d}g_{\mathrm{syn}}(t)}{\mathrm{d}t} = -g_{\max} (t - S_{\mathrm{syn}}(t)) \exp(-(t - S_{\mathrm{syn}}(t)) - \tau_{\mathrm{syn}}) \quad (3)$$

where syn $\in$ {exc, inh}, $S_{\mathrm{syn}}(t) = \delta(t - tk)$ are the k incoming synaptic spike trains where $i \in \{1,\ldots,n\}$ refers to presynaptic neurons and k to the different spike times of these neurons. The synaptic time constants were chosen to be $\tau_{\mathrm{exc}} = 3$ ms and $\tau_{\mathrm{inh}} = 7$ ms for excitation and inhibition, respectively. The reversal potentials were $E_{\mathrm{exc}} = 0$ mV and $E_{\mathrm{inh}} = -80$ mV. The synaptic weights $g_{\mathrm{syn}}(t)$ were set to $g_{\mathrm{exc}} = 1$ nS for the excitatory conductance, and a balance of $g_{\mathrm{inh}} = 5$ nS unless stated otherwise. We used a constant delay of 0.5 ms, equal for all connections.

Network architecture

We simulated a microscope focal plane area of 100 × 100 µm as a two-dimensional, layer-like network (with periodic boundary conditions to avoid size effects). Neurons were arranged on a grid meant to represent ~10-µm separation between neurons. Every neuron was sparsely connected with the rest of the network with a connection probability drawn from distance-dependent probability distribution. Each neuron i was connected with another neuron j with a probability p_{ij}, depending on their distance r_{ij}, according to an exponential profile:

$$p_{ij} = A_{\mathrm{syn}} \times \exp^{-(\lambda_{\mathrm{syn}} r_{ij})} \quad (4)$$

where A_{syn} is the amplitude of the exponential, varying the number of neurons sampled in the chosen radius r_{ij}. The amplitudes A_{syn} were set to have the number of connected neurons that roughly match the distributions previously described[82,85,86], considering the topological distribution and the excitatory/inhibitory neuron ratio ($A_{\mathrm{exc\text{-}exc}} = 14$, $A_{\mathrm{exc\text{-}inh}} = 24$, $A_{\mathrm{inh\text{-}exc}} = 14$, $A_{\mathrm{inh\text{-}inh}} = 24$), resulting in a degree of ~75 connections per neuron. The decay controller λ_{syn} was set to $\lambda_{\mathrm{exc}} = 1.2$ and $\lambda_{\mathrm{inh}} = 1.5$. The seed to the random number generator responsible for the sampling of the connections was systematically varied, producing $n = 20$ different networks, but following the same type of exponentially decaying distributions. In the tests for the relationships between distance-dependence and flow measures, we systematically varied the amplitudes, lambdas (and correspondingly synaptic weights to keep the conductance balance constant across network configurations). The arrangement of connections was tested for changes in synaptic weights. See the summary table of modularity measures and population events in the model (Supplementary Table 2).

Drive

An external population of Poisson drivers was connected to both the regular spiking and fast spiking populations, with uniform random probability drawn from the half-open interval (0,0.01), with a fixed synaptic weight of 0.5 nS.

Simulator

All simulations were performed using the NEST simulator[87] controlled through the PyNN interface[88].

Comparison of model and MICrONS properties

Simulating a network whose only structure arose from distance-dependent connectivity yielded properties comparable to the MICrONS dataset, listed below (Supplementary Table 3).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets analyzed during the current study are available in the CodeOcean capsule hosted by Nature (https://doi.org/10.24433/CO.9782876.v3).

Code availability

The custom code used to preprocess data, conduct analyses and reproduce all figures is available in the CodeOcean capsule (https://doi.org/10.24433/CO.9782876.v3).

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Author contributions

D.G. and A.D. conceptualized the study, performed the investigation and contributed to writing, reviewing and editing the final draft of the paper. D.G. and A.F. performed the methodology and validation. D.G. performed the software development and visualization, and wrote the original draft of the paper. A.D. was responsible for funding acquisition, project administration and supervision.

Competing interests

All authors declare no competing interests.

Additional information

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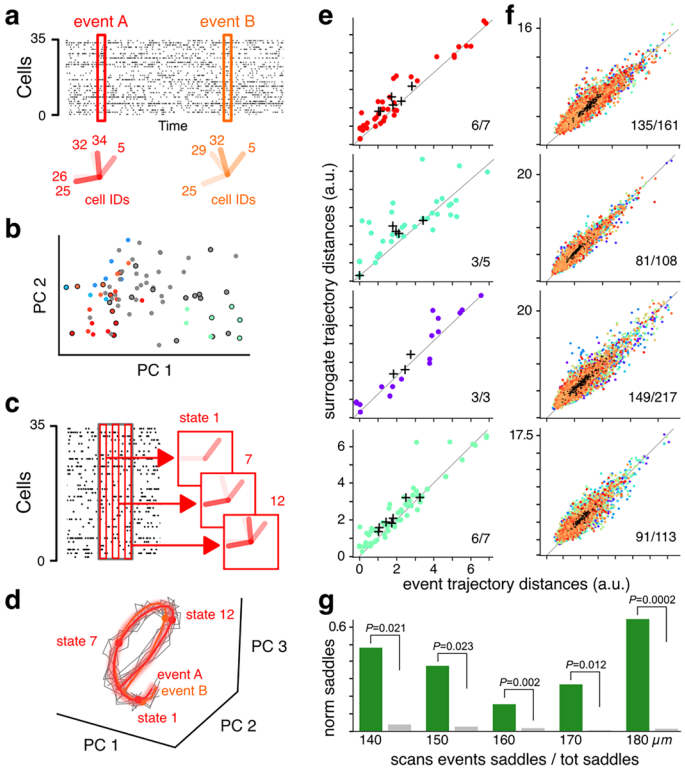

Extended Data Fig. 1 | Reproducible patterns are transient attractors. a, Top, population events were identified (as discussed in Fig. 1, two examples A and B). Bottom, the dF/F value of each cell participating in the event can be represented as a radial multidimensional vector defining the state of the system. **b**, An eigenvalue decomposition allows the choice of a suitable reduced set of axes to represent the multidimensional vectors. By looking at their (reduced) state space, the events are grouped in space and colored as the clusters in Fig. 1b (those occurring during stimulus presentations have black edges). Their Silhouette distance confirmed the clustering (see text). **c**, We can zoom in to each event dividing it into smaller states (left). Each state is characterized by cell dF/F vectors as in **a** (right). **d**, The set of states of an event forms a trajectory in the (reduced) state space (red and orange lines). To establish a significant distance, we created surrogate trajectories (gray lines) by shuffling cell vectors, and we measured the multidimensional (Hausdorff) vector distance between population event trajectories against that of surrogate trajectories. **e**,**f**, Each event is represented by a dot, colored by its cluster. Its coordinates are the Hausdorff distances within the cluster and within surrogates (black cross at mean trajectory distance, mean surrogate distance). The majority of cortical recordings presented shorter distances (points colored by cluster, with means as crosses) between event trajectories compared to surrogates (**e**, events from the MICrONS dataset, **f** events from the Neuropixels dataset, with some confidence intervals shown on the right column of the associated notebook). **g**, In the MICrONS data, where the ΔF/F data were made available, the proportion of saddle node-contributing neurons overlapping with known cluster cores was calculated for each scan (5 scan depths in μm). Core neurons (green) occurred significantly more inside saddle node events in all scans.

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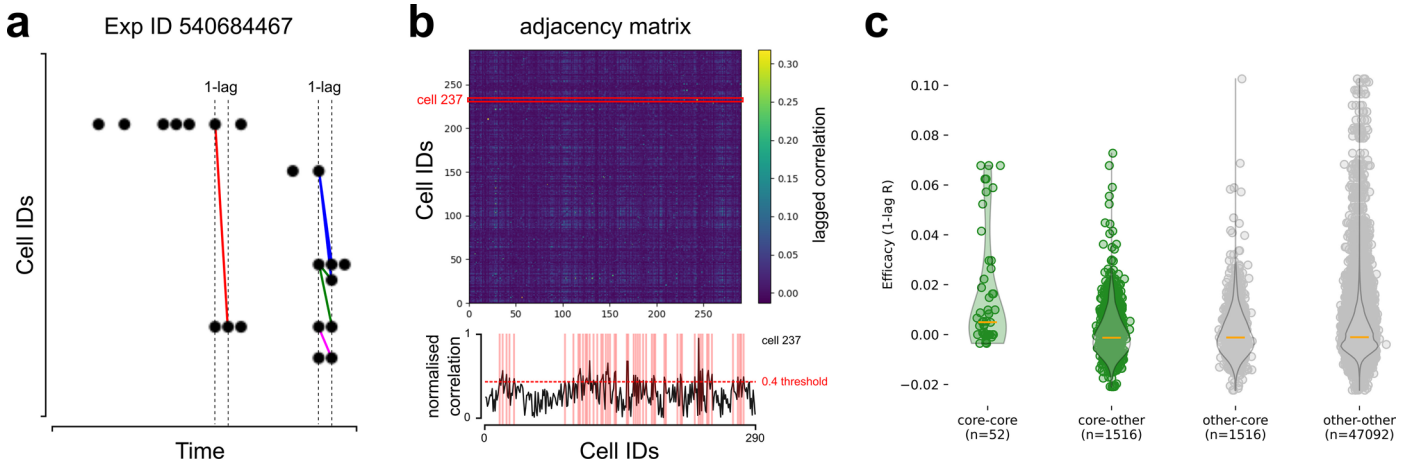

Extended Data Fig. 2 | Core functional correlation. a, Detail from the Allen Brain Observatory experiment ID 540684467 raster plot showing calcium spikes (black circles). Each row contains the calcium spikes for a cell. The minimal interval between two spikes is the two-photon imaging time resolution (30 Hz for the Allen Brain dataset). This interval represents the 1-lag correlation interval to identify cells potentially connected. In the figure, only some cells with 1-lag correlated firing were joined by colored segments to improve legibility. **b**, The vector of 1-lag correlations for each cell is collected into a matrix (top). Weak correlations (threshold = 0.4) were not considered for further analysis (bottom, correlations for cell ID 237). **c**, Core neurons are those sharing (nonsignificant, t-score = 1.53, $p = 0.12$) more high functional correlation across events compared to others.

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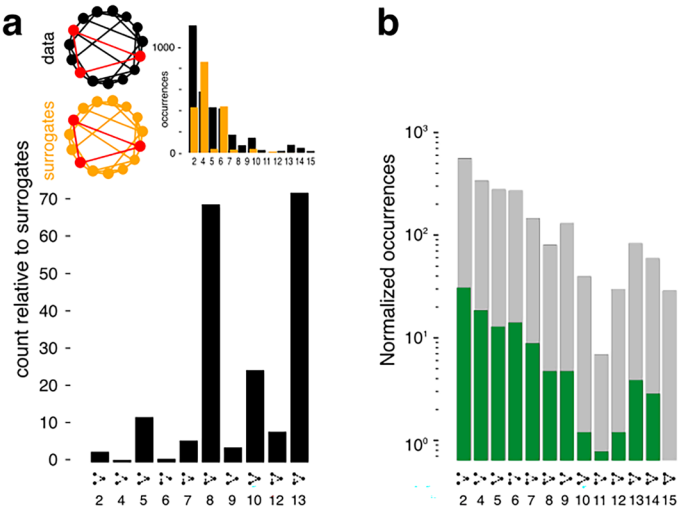

Extended Data Fig. 3 | No specific motifs for core neurons. We looked at the distribution of motifs—statistically significant connectivity patterns considering groups of three cells. To understand the significance of motif occurrences, we used the ratio of real vs 100 surrogate degree-matched networks. **a**, Connectivity motifs distribution for all two-photon recorded neurons (top, occurrences count). The ratio followed a known distribution showing an abundance of three-cell mutual motifs. **b**, Normalized motif occurrence distribution for cores and other neurons. Both made a mixture of mutual and nonmutual connections.

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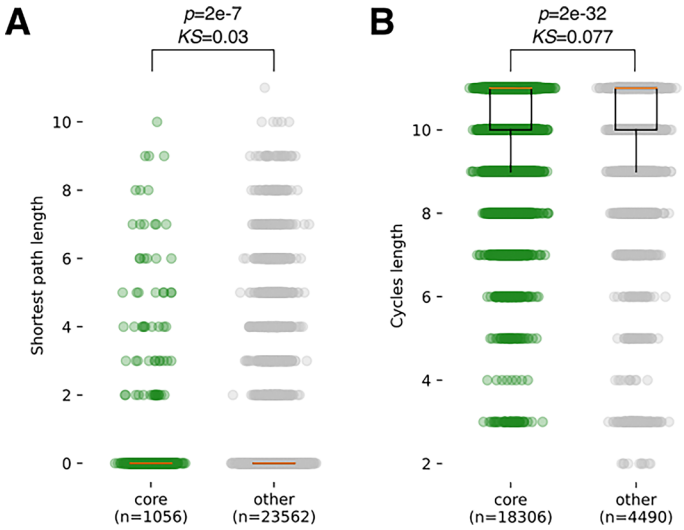

Extended Data Fig. 4 | Cores are not recursively connected via multisynaptic feedback. Even if cores were not directly connected, they could be highly connected via secondary paths, abundant enough to ensure that they are pattern completion units in traditional attractor network terms. **A**, Core-to-core shortest paths were neither more nor shorter than other-to-other paths (0.27 ± 1.19 vs 0.16 ± 0.92 length, Kruskal–Wallis t = 26.4, p = 2.7 ×10^{-7}, KS = 0.03). **B**, Core neurons could be part of looped paths, circling back to them, and providing a weak form of recursion. Core-based cycles were more (green, 18,306) compared to other-based cycles (gray, 4,490). But core and noncore cycles had on average the same length (10.34 ± 1.07 and 9.93 ± 1.76 connections, with minimal effect size KS = 0.077), same as the network diameter (d = 10).

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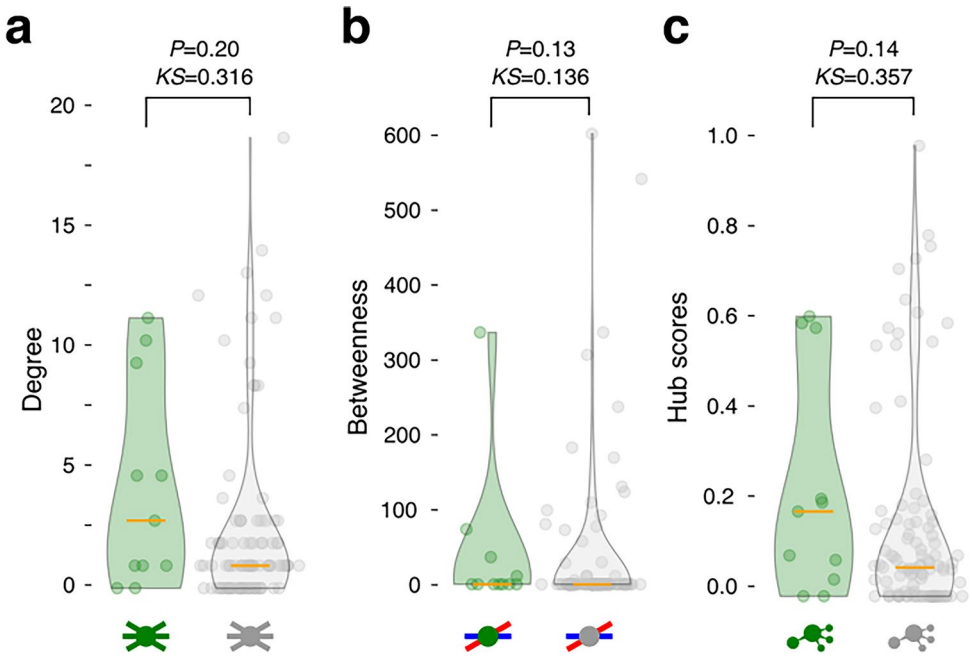

Extended Data Fig. 5 | Simple centrality measures show no difference between cores and other neurons. a, The degree—number of connections per neuron—of cores was not significantly higher than other neurons. **b**, The betweenness—the number of shortest paths between any two nodes passing by a considered node—of cores was not significantly different from other neurons. **c**, The hub score—the weighted number of outward connections of a node that points to central nodes—of cores was not significantly higher than other neurons.

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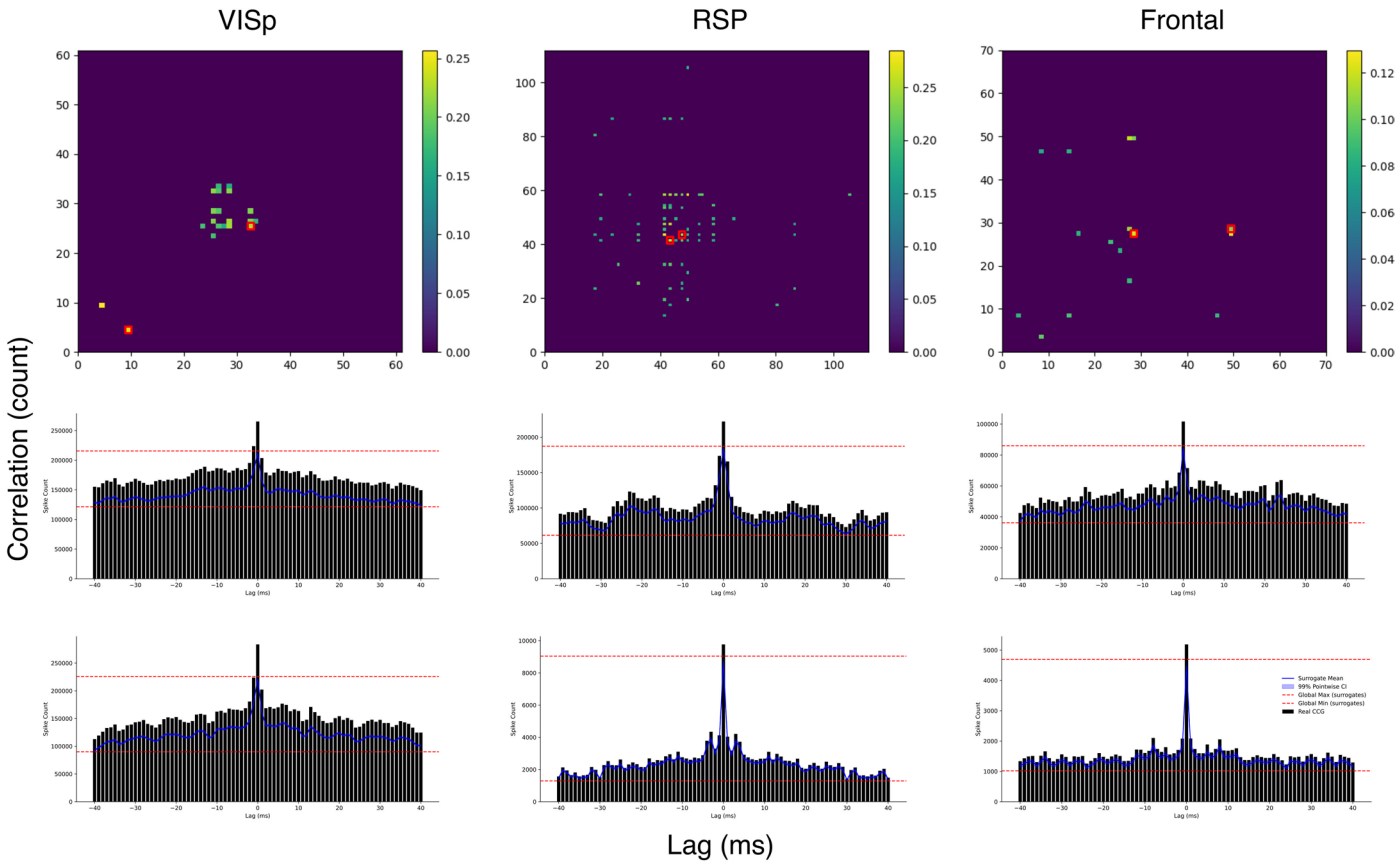

Extended Data Fig. 6 | Cross-correlograms of Neuropixels units. Cross-correlation was computed over correlated units from the Neuropixels datasets for the areas of interest (top, population correlograms, red squares are the units correlograms shown below). Correlograms were computed as in ref. 89. Briefly, spike trains from pairs of units were convolved using bins of 2 ms over a window of 40 ms (black bars). Then, each spike train was independently jittered within ±5 ms to generate 1,000 surrogate datasets. The surrogate mean (blue curve), 99% pointwise (blue shaded area) and global confidence bands (red lines) are derived from the surrogate distributions. This analysis informed our choice of the time window (1–5 ms) to estimate monosynaptic connections in our functional connectivity analysis (Methods).

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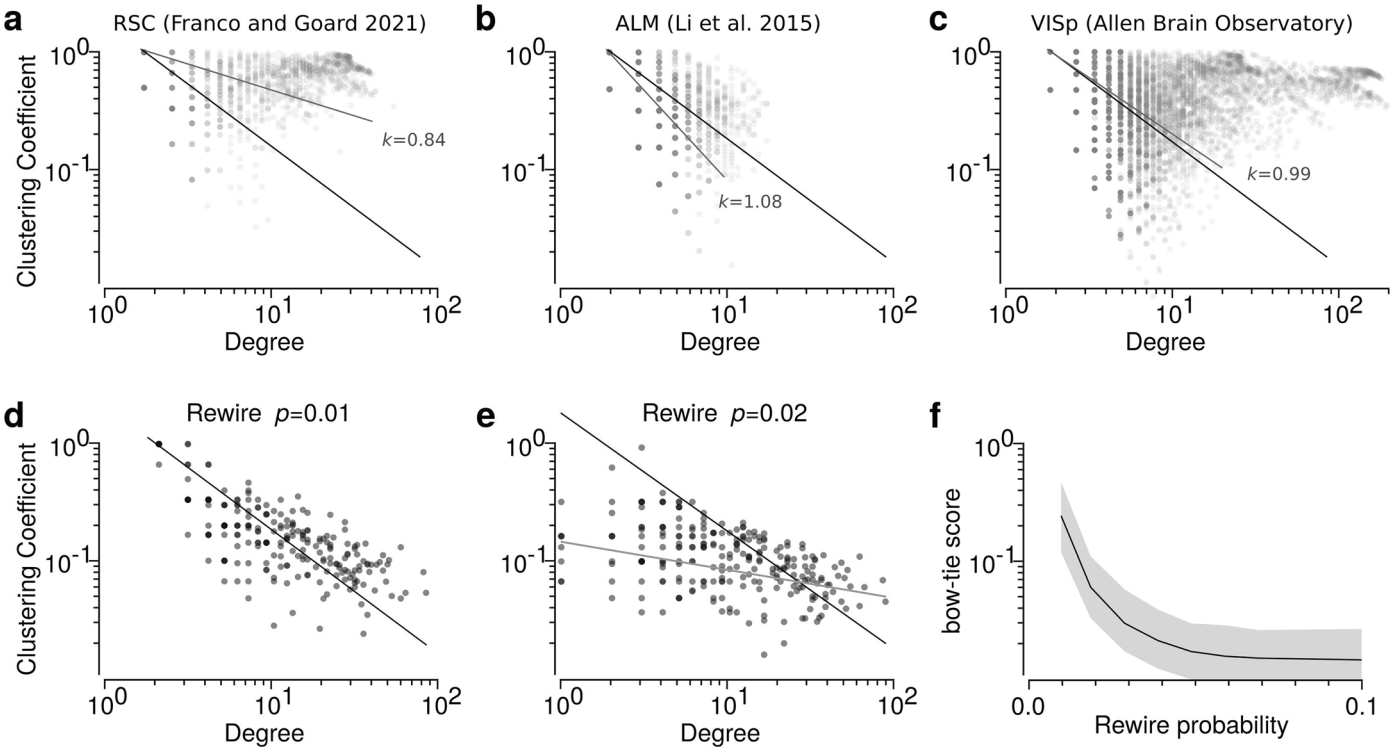

Extended Data Fig. 7 | Hierarchical modularity across cortices and rewiring of the MICrONS dataset. a–**c**, Example hierarchical modularities for other cortices (retrosplenial cortex, from ref. 32, and anterolateral motor cortex, from ref. 33), computed from functional activity using the method discussed in ref. 38. Each cortical region had a different slope (in black the fitting curve for the MICrONS data). This modularity, estimated using functional connectivity, may tend to overestimate large degrees/large clustering coefficient, as evident comparing the same area (VISual primary) functional hierarchical modularity of Fig. 3f and h. **d**,**e**, Example hierarchical modularity resulting from rewiring the MICrONS graph. **d**, rewiring the graph edges with probability p = 0.01 results in a hierarchically modular relationship as that observed in the data. In **e**, already at p = 0.02 the relationship deteriorates (fitting curve in black is kept in both panels). **f**, The bow-tie score (ratio of modules with clearly identifiable submodules converging-diverging from a central module over the total number of modules) rapidly deteriorates for increasing rewiring probabilities (s.e.m. reported as gray shaded area).

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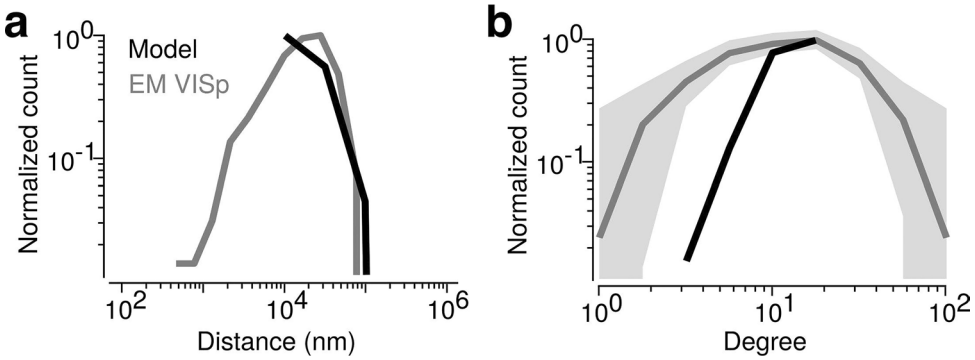

Extended Data Fig. 8 | Comparison of the MICrONS and model basic network properties. a, Distribution of distances between connected neurons in the MICrONS dataset (gray). Notice that its shape follows the ones found in refs. 76,82,85. We fitted our model (black) to the data, considering the space available given the intercell minimal distance assumed as model spatial and density resolution (1 cells/100 μm^2, for a total of 10,000 excitatory cells per mm^2, 2,500 inhibitory cells are distributed accordingly to span the same space). **b**, Normalized count of degrees in the MICrONS dataset (gray, with shaded s.e.m.). In the model (black), the max degree is 75, which we imposed in all models. For small distance-dependent connectivity ranges (from 22 to 42 μm) the chosen intercell minimal distance resulted in a lower number of connections in the model.

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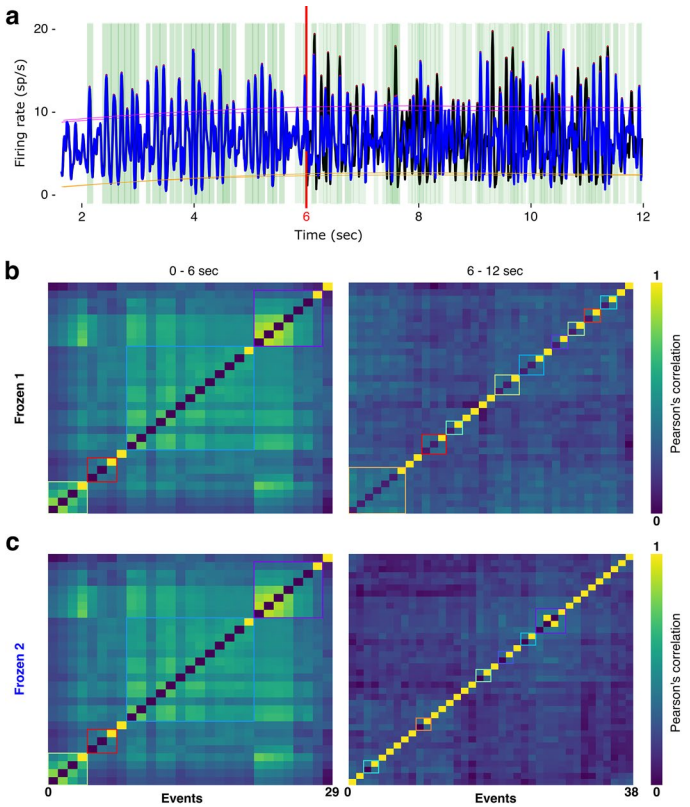
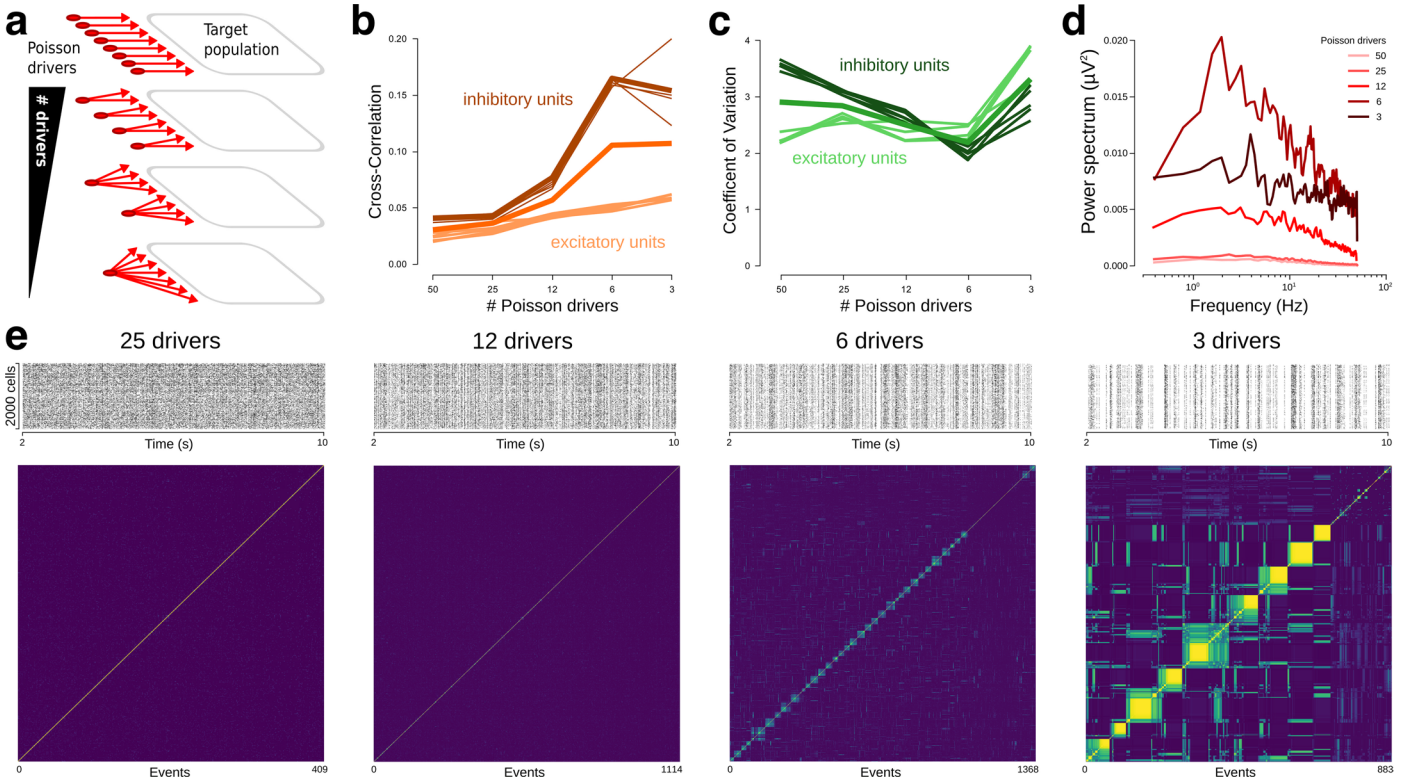


Extended Data Fig. 9 | Specific clusters for different stimuli. We tested the ability of our mechanistic model to support pattern completion to show that it can recapitulate phenomena in V1 that have been specifically cited as evidence that it is an attractor network. In many previous models, fixed-point attractors (pattern completion subnetworks) are achieved either through learning or artificially imposed. In our model no learning nor artificial subnetwork was imposed, yet reproducible firing patterns are observed. We used a 'frozen noise' input strategy to see whether it leads to same pattern completions. Three spike train matrices (called A, B, C, not shown) were generated using Poisson populations with different seeds (all having on average 5 sp/s). We then created two spike trains matrices ('frozen 1' and 'frozen 2') in which the first half was identical for both (spiketrain A), and the second half varied (either spiketrain B or C). Then we ran the same model as in main Fig. 4 using the two (frozen) spike train matrices to test whether it reproduced the same events during the identical first half, and different events in the second half. **a**, Firing rates from the network stimulated with the two spike train matrices ('frozen 1', black, and 'frozen 2', blue). The first portion of the firing (up to 6 sec, red vertical line) is the same for both runs. The second portion of the firing rate was different (green vertical shades: synchronous events). **b**,**c**, The clustering of population events shows the complete reproducibility of activity for the initial portion (left, **b** and **c**). Different clusters emerged in the second portion of the stimulus (right, **b** and **c**) (colored squares: clusters of vectors sharing neurons beyond a surrogate-based threshold).

https://doi.org/10.1038/s41593-025-02128-5



Extended Data Fig. 10 | Increasing input correlations rescues population event reproducibility but introduces network-wide oscillations. a, For the same target population (right), we systematically reduced the number of Poisson input drivers (red disks and arrows), to explore how the increase in input correlations affected the correlations between population events. **b**, Reducing the number of Poisson input drivers (n = (50, 25, 12, 6, 3)) increased the population cross-correlation of spikes toward values characteristic of oscillating regimes (from CC = 0.035 at n = 50, to CC = 0.11 at n = 3). **c**, The number of Poisson drivers did not significantly alter the population coefficient of variation. **d**, Reducing the number of Poisson input drivers increased the power for slow oscillations in the Fourier spectra of the firing rates. **e**, Four 10-s spike rasters for 2,000 example cells, with their correlation matrices, show the changes in firing regimes responsible for the increase in population events correlation. The first correlation matrix on the left shows the absence of reproducibility. In the right correlation matrix, the off-diagonal correlations show the presence of correlations between population events, resulting in higher values of reproducibility.

nature portfolio

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Last updated by author(s): Sep 18, 2025

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Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

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Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See Eklund et al. 2016)

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☐	☒ Graph analysis
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Functional and/or effective connectivity	Pearson's correlation
Graph analysis	Adjacency matrix built by coordinating pre- and post- soma root IDs from the MICrONS project. All graph analyses were performed using the Python version of the library iGraph: degree, clustering coefficient, information flow, pagerank, shortest path, betweenness, hub score
Multivariate modeling and predictive analysis	PCA